## 1. Title

# Centralized Priority Resolution with Dynamic Preemption Transfer for Concurrent Emergency Vehicles in Software-Defined Traffic Networks

## 2. Abstract

Efficient traffic-light preemption is essential for reducing emergency vehicle response time. However, existing Software-Defined Traffic Light Preemption (SD-TLP) systems have mainly been evaluated for single-emergency-vehicle scenarios, while existing multi-emergency-vehicle approaches generally rely either on decentralized intersection-local rules or complex centralized optimization methods. This work addresses this gap by extending the centralized SDN-based traffic-light preemption to support concurrent emergency vehicles through vehicle-type and estimated-arrival-time (EAT)-based priority rules together with a proposed dynamic preemption transfer mechanism that cancels an active preemption and reassigns it to a higher-priority emergency vehicle when it becomes eligible. The proposed approach was evaluated in OMNeT++ under four scenarios involving one to three emergency vehicles across low, medium, and high traffic conditions using ten random seeds per configuration. The evaluation considers both emergency vehicle performance and the impact on normal traffic using Average Rescue Time (ART), Average Waiting Time (AWT), Average Imposed Waiting Time (AIWT), controller overhead, and transfer-related characterization metrics. The results show that the proposed approach generally reduces emergency vehicle rescue time while maintaining an acceptable impact on normal vehicles compared with the baseline approaches, particularly under moderate and high-traffic scenarios. Under Scenario 2 with medium traffic, the proposed approach achieved an average ART of 59.65 s compared with 95.25 s for SD-TLP with FCFS and 135.15 s for No Preemption. The transfer characterization results also indicate positive net rescue-time gains under most low and medium-traffic conditions. However, the results also indicate that under high traffic conditions, the dynamic preemption-transfer mechanism is not uniformly beneficial, particularly in cases involving two emergency vehicles. Overall, the findings show the traffic conditions and scenarios in which centralized multi-emergency-vehicle preemption and dynamic preemption transfer are beneficial, and the conditions in which they are not. Comparison against decentralized intersection-local approaches is left for future work.

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# 3. Introduction

Efficient traffic management remains a major challenge in modern urban environments due to increasing traffic congestion and vehicle density. Congested intersections can significantly delay emergency vehicles such as ambulances, fire trucks, and police vehicles.

Traffic-light preemption systems aim to reduce emergency vehicle delay by temporarily modifying normal traffic-signal operation to prioritize emergency vehicles at intersections. Different approaches have been proposed for emergency vehicle traffic-light preemption, including decentralized intersection-local control, queue-aware preemption strategies, and centralized software-defined architectures. One such approach is the Software-Defined Traffic Light Preemption (SD-TLP) mechanism proposed by Bagheri et al. [1], which uses a centralized controller to dynamically manage traffic signals along the route of an emergency vehicle. However, most existing traffic-light preemption approaches, including SD-TLP [1], mainly focus on scenarios involving a single emergency vehicle. In real environments, multiple emergency vehicles may simultaneously request traffic-light preemption while approaching the same intersection from different directions. These situations can create conflicts between preemption requests, which lead to increased rescue time and inefficient traffic-signal operation if they are not properly managed.

Existing studies that address multiple emergency vehicles generally follow two main approaches. The first approach uses decentralized intersection-local decision making based on simple priority rules, such as emergency vehicle type and estimated arrival time (EAT), as done by Younes and Boukerche [2] and Humagain and Sinha [3]. The second approach relies on centralized optimization methods such as fuzzy-based control and mixed-integer linear programming (MILP) [4, 5]. Although these optimization-based approaches can improve emergency vehicle coordination, they often introduce high computational complexity and increased runtime overhead.

In contrast, centralized rule-based approaches for multi-emergency-vehicle traffic-light preemption have received limited attention in the published literature. To our knowledge, centralized rule-based traffic-light preemption within the SD-TLP system has not been evaluated under concurrent multi-emergency-vehicle conditions. In particular, limited work has investigated centralized rule-based mechanisms that dynamically re-evaluate and transfer active traffic-light preemption sessions when a higher-priority emergency vehicle becomes eligible.

To address this gap, this work investigates centralized rule-based multi-emergency-vehicle traffic-light preemption within the SD-TLP system. The proposed approach introduces a dynamic preemption-transfer mechanism that re-evaluates active preemption sessions and transfers priority to a higher-priority emergency vehicle when it becomes eligible. In addition, this work characterizes the dynamic preemption-transfer mechanism by analyzing transfer frequency and its impact on emergency vehicle rescue time under different traffic conditions.

The main contributions of this work are summarized as follows:

- Extending the centralized SD-TLP framework to support concurrent emergency vehicles using rule-based conflict resolution.
- Introducing a dynamic preemption-transfer mechanism that re-evaluates active preemption sessions and transfers priority to higher-priority emergency vehicles when necessary.
- Experimentally characterizing the dynamic preemption-transfer mechanism by analyzing transfer frequency and its effect on emergency vehicle rescue time.
- Identifying the traffic conditions in which centralized multi-emergency-vehicle preemption is beneficial, as well as conditions where its coordination overhead reduces its effectiveness.

To evaluate the proposed approach, OMNeT++ is used to simulate four scenarios involving one to three emergency vehicles under low, medium, and high traffic conditions. The proposed method is compared against SD-TLP with FCFS, SD-TLP Closest, and No Preemption using metrics including Average Rescue Time (ART), Average Waiting Time (AWT), Average Imposed Waiting Time (AIWT), and transfer-related characterization metrics.

The remaining part of this report is organized as follow: Section 4 presents the necessary background on Vehicular Ad Hoc Networks (VANETs), Software-Defined Networking (SDN), and traffic-light preemption. Section 5 reviews related work on traffic signal control for both single and multiple emergency vehicles and discusses the identified research gap. Section 6 presents the proposed centralized rule-based multi-emergency-vehicle SD-TLP mechanism, including the system overview, the original SD-TLP mechanism, and the proposed dynamic preemption-transfer mechanism. Section 7 describes the experimental setup, including the simulation environment, evaluated scenarios, baselines, performance metrics, and experimental design. Section 8 presents and discusses the experimental results, including the characterization of the dynamic preemption-transfer mechanism and ethical considerations. Section 9 discusses the threats to validity and limitations of the study. Finally, Section 10 concludes the report and summarizes the main findings.

# 4. Background

This section provides the background information necessary to understand the concepts discussed in this project, including an overview of Vehicular Ad Hoc Networks (VANETs) [6], Software-Defined Networking (SDN) [7], and traffic light preemption.

## 4.1 Vehicular Ad Hoc Networks (VANETs)

Vehicular Ad Hoc Networks (VANETs) are vehicle communication networks that are designed to enable communication among vehicles and between vehicles and roadside infrastructure, such as traffic lights and roadside units (RSUs) [6]. By supporting traffic management and driving assistance, VANETs play a key role in intelligent transportation systems. Generally, VANETs consist of vehicles equipped with on-board units (OBUs), which are hardware elements that allow wireless communication with other OBUs or RSUs [8]. While RSU is an infrastructure device generally used for data transmission in a VANET environment, such as between vehicles and a central controller [9].

Moreover, based on the entities involved, VANET communication can be classified into several types. These types include vehicle-to-vehicle (V2V) communication, where vehicles exchange information directly, such as accident warnings, vehicle-to-infrastructure (V2I) communication, where communication occurs between vehicles and roadside infrastructures, most commonly RSUs, and infrastructure-to-infrastructure (I2I) communication, where infrastructures communicate with each other by sharing data such as traffic density [6].

To support these communication types, VANETs employ several wireless technologies. Early deployments of VANETs primarily rely on Dedicated Short-Range Communications (DSRC) [6], which is a short-range wireless communication technology that operates in the 5.9 GHz frequency band and is standardized under IEEE 802.11p, enabling V2V and V2I communication [10]. However, with the expansion of VANET applications, additional technologies such as Bluetooth communication and satellite communication have also been explored.

Through these communication capabilities, VANETs support a wide range of applications and services, such as road blockage warnings, traffic congestion information, and traffic signal control, contributing to improved traffic efficiency and road safety [6].

In this project, the focus is mainly on Vehicle-to-Infrastructure (V2I) communication, where emergency vehicles communicate with RSUs to help enable traffic light preemption.

### 4.2 Software Defined Networking (SDN)

Traditional computer networks are built with both the control plane and data plane integrated within the same devices such as routers and switches. However, Software-Defined Networking (SDN) [7] introduces a different approach by separating these two planes. This separation enables network control to be centralized in a software-based controller, making the network easier to manage and more flexible. In SDN, network devices are responsible only for forwarding traffic, while decisions about how that traffic flows are handled centrally [7].

Communication within the SDN architecture is achieved through interfaces. The southbound interfaces allow the controller to communicate with the forwarding devices and manage them by installing flow rules and collecting network information. On the other hand, the northbound interfaces enable communication between the controller and the application layer which allows developers and network administrators to define network behavior, policies, and services through the software [7].

The SDN architecture is organized into three main layers. The application layer contains network applications and services such as routing and security. The control layer includes the SDN controller which acts as the central point of decision-making, while the infrastructure layer consists of physical or virtual switches and routers responsible for forwarding packets [11].

In this project, we use an SDN-based centralized controller that has a global view of traffic conditions to make traffic light preemption decisions. These decisions will then be implemented by the traffic light.

### 4.3 Traffic Light Preemption

Intelligent Transportation Systems (ITS) [12] refer to transportation systems that apply communication, sensing, and information technologies to improve traffic efficiency, safety, and transportation management. These systems combine different technologies, such as sensors, cameras, and communication networks, to collect and exchange traffic information in real time. By analyzing this information, ITS can help monitor traffic conditions, detect congestion or accidents, and support better traffic control decisions across modern transportation networks [12].

One important application of ITS is traffic light preemption, which temporarily changes the normal operation of traffic signals to give priority to specific vehicles or movements [13]. In most cases, it is used for emergency vehicles such as ambulances, fire trucks, and police vehicles [14]. Traffic light preemption helps reduce the response time of emergency vehicles by allowing them to pass intersections with minimal delay [14]. This is usually achieved by extending the green phase or shortening the red phase for the direction of the emergency vehicle so that it can safely cross the intersection. In modern ITS environments, traffic light preemption can also be implemented using communication technologies such as vehicle-to-infrastructure (V2I) communication, where emergency vehicles send information about their location and speed to nearby traffic signal controllers.

Although traffic light preemption improves emergency response, it can increase delays for normal vehicles if it is not carefully managed [14]. For example, interrupting the normal signal cycle may create longer queues or temporary congestion for vehicles traveling in other directions. Therefore, recent research focuses on adaptive and non-intrusive preemption strategies that balance emergency vehicle priority with overall traffic flow by properly managing signal timing and the recovery process after preemption [1]. These strategies aim to ensure that emergency vehicles receive priority while minimizing the negative impact on regular traffic operations.

## 5. Literature Review

This section provides a comprehensive review of existing work, focusing on traffic signal priority mechanisms for scenarios involving a single emergency vehicle (EV), as well as few studies that address traffic signal priority for multiple emergency vehicles.

### 5.1 Traffic Signal Control for Single Emergency Vehicle

This subsection highlights a review of traffic light control mechanisms and algorithms that are designed to grant priority to a single emergency vehicle. A summary of these works can be found in Table 2.

Bagheri et al. [1] proposed a Software-Defined Traffic Light Preemption (SD-TLP) to reduce Emergency Medical Vehicle (EMV) arrival time while limiting disruption to normal traffic using a centralized controller. Their framework also selects the nearest emergency center and the optimal route to the accident location using Dijkstra's algorithm [15]. In addition, they include a lane-changing strategy in which nearby vehicles change lanes after receiving notifications from the EMV. The SD-TLP algorithm determines when and where preemption of signals should start, observes the next two traffic lights on the EMV route to decide whether they should preempt under heavy traffic, and restores normal operation through a recovery strategy after the EMV passes. The framework was evaluated using OMNeT++ [16], SUMO [17] , and VEINS [18] simulation tools on a map of Tabriz, Iran, using Average Rescue Time (ART), Average Imposed Waiting Time (AIWT), and Average waiting Time (AWT) as performance metrics across several scenarios. The results show that SD-TLP provides the best balance by reducing ART and maintaining acceptable AIWT and AWT levels. However, the authors suggest that further improvements can be made by evaluating the proposed approach with a larger number of vehicles that have different priorities.

Zlatkovic et al. [19] proposed a connected vehicle (CV) based algorithm that preempts traffic signals when an emergency vehicle is within 600 ft of a signalized intersection. When the emergency vehicle is in the detection range, it sends its location, route, and speed to the traffic signal controller through the roadside unit (RSU). The controller then switches the traffic signal in the emergency vehicle's approach to green, clearing vehicles ahead and allowing the emergency vehicle to cross the intersections with minimal delays. The approach was evaluated using VISSIM microsimulation [20] on a map consisting of ten signalized intersections and assessed under multiple traffic scenarios. The results show that the proposed algorithm significantly improves emergency vehicle delays by up to 36% and increases its travel speed by up to 50%, while having minimal impact on regular vehicle traffic.

Moreover, Rego et al. [21] proposed a Software Defined Network (SDN) and Internet of Things (IoT)-based architecture for emergency traffic management in smart cities. The system connects traffic cameras and signals through OpenFlow switches managed by an SDN controller. This SDN controller coordinates emergency response by identifying the emergency type, allocating resources, computing optimal routes, and dynamically adjusting traffic signals to prioritize emergency vehicles. The proposed approach was evaluated using Mininet [22], where results show a reduction in network delay of approximately 33% for emergency traffic and 50% for normal traffic. The authors also demonstrate that energy consumption increases linearly with the number of IoT nodes, indicating good scalability.

The study in [23] proposed a traffic signal control system, based on VANETs and V2I communication, for prioritizing emergency medical vehicles (EMVs) while minimizing the delay for regular vehicles. The system continuously receives vehicle and intersection data to detect the presence of an EMV and estimates its arrival time at the upcoming intersection. If the estimated arrival time is lower than a predefined threshold, traffic signal timings are adjusted to prioritize the EMV. In addition, the system also optimizes the traffic signals based on the current traffic conditions. Experimental results show that the proposed method achieves positive optimization performance across all traffic conditions when compared with several other methods.

Chen et al. [24] proposed an Emergency Vehicle (EV) Traffic Signal Preemption that considers queue spillbacks along the routes of an EV while also taking into account the negative impacts of signal preemption on non-priority traffic. The proposed system manages queue length before the EV departs from the depot to ensure that the queues are cleared, activates signal preemption based on EV's estimated arrival time at intersections, and handles queue spillback occurring from downstream intersections by activating signal preemption at downstream intersections first and then moving upstream. Additionally, because signal preemption can negatively impact normal traffic, the system also includes a recovery mechanism. During peak hours, the system reduced the travel time of emergency vehicles by up to 60% when compared to normal operations and by 31–41% when compared to an existing preemption strategy [25].

Additionally, Shaaban et al. [26] proposed an emergency vehicle preemption (EVP) method that consists of two main parts: a preemption strategy and a best-path selection strategy. First, the preemption strategy aims to minimize both the traffic facing the emergency vehicle (EV) and normal traffic by starting the preemption at the correct time, which is based on the average queue length and queue discharge speed. The second part is the best-path selection, which calculates the minimum travel time using Dijkstra's algorithm [15] while taking into consideration the location of both the EV and the incident. The proposed method was tested using VISSIM [20]. The results show that the travel time for EVs in traffic conditions was reduced by 15.8% to 48.8%, and the average travel time for an EV was reduced by 24.3%, showing that the proposed EVP method is beneficial for EVs.

Zhong et al. [27] introduced a traffic signal control strategy to prioritize emergency vehicles based on three main steps: on-demand signal timing, a signal preemption strategy, and a recovery cycle mechanism. Their system uses a decentralized intelligent agent (AGT) at each intersection to manage signal operations, applying non-intrusive preemption or intrusive preemption depending on whether the emergency vehicle can pass within the available green time. A recovery strategy is then used to discharge queued vehicles based on the queue length. The simulations were conducted using SUMO [17]. The proposed strategy achieves a travel time reduction of 62.85% when compared with the fixed-time control method (FTCM), 50.83% compared with Min's flexible signal preemption method (FSPM) [28], and 11.62% compared with Qin's intrusive signal preemption method (ISPM) [29].

Furthermore, Rosayyan et al. [30] proposed a control strategy for traffic signals that allows EVs to pass intersections quickly while reducing the waiting time for normal traffic. In this study, they assume the city is split into different zones, and each zone is assigned an edge server. In addition, their system uses a GPS based IoT sensor as a Dashboard Unit

(DBU). Every 1 s, the DBU sends the Location Information (LI) of the EV to the closest edge server to resolve traffic ahead of it. From the information sent to it, the edge server calculates the Time to Reach the Junction (TRJ) based on the average speed and distance of the EV. Based on this estimation, traffic signal timings are dynamically adjusted to allow EVs to pass the intersection and then restore the original signal timing. The results show that the proposed strategy reduces the waiting time for normal vehicles by 73.23% when compared with existing systems.

Similar to the previous approaches, the study in [31] developed a Dynamic Preemption Algorithm aimed at reducing the travel time of a single emergency vehicle (EV) while minimizing traffic disturbances at signalized intersections. Their approach uses Vehicle-to-Infrastructure (V2I) communication to obtain the EV's data and estimates its arrival time to the intersection, which triggers one of two control strategies: Green Extension or Red Truncation. They also use a Queue Dissipation logic, which analyzes the traffic density ahead of the EV and activates the green signal early enough to clear the queue. After the EV clears the intersection, a Recovery Phase is initiated to restore signal stability. The proposed method was evaluated using VISSIM microsimulation [20], demonstrating a significant reduction in EV intersection delays and stops compared to both fixed-time control and conventional static preemption. However, the approach relies on robust V2I communication and only considers scenarios involving only a single EV.

Continuing with signal prioritization for a single emergency vehicle, Min et al. [28] proposed an on-demand green wave system to reduce the response time of emergency vehicles in road networks. Their approach focuses on reliable route planning and traffic signal preemption. First, the routing method selects paths by considering both the average estimated time of arrival and its variation. Second, the authors introduced a non-intrusive signal preemption strategy so the emergency vehicle can pass intersections without stopping while considering normal traffic. The proposed system was evaluated using data-driven simulations and real-world test conducted on actual roads in Hangzhou, China. Simulation results showed that the proposed routing method achieves similar average travel time to traditional shortest-path methods but with much lower travel time variation. Real-world results showed that the signal preemption strategy reduced the emergency vehicle response time by 30%, while delay for normal vehicles were not significantly affected.

Table 2: Summary of Traffic Signal Control for Single Emergency Vehicle

| Ref. | Citation | Focus | Domain | Tool/ Simulator | Baselines | Performance measures | Notes (strength/limitation) |
|---|---|---|---|---|---|---|---|
| [1] | N. Bagheir et al., 2024 | To minimize the arrival time of Emergency Medical Vehicles to the accident location while limiting disruptions to normal city traffic | VANET, Smart Cities, Intelligent Transportation System | OMNeT++, SUMO, Veins | Normal Traffic, Local Preemption, Full Preemption, and lane-changing | Average Waiting Time (AWT), Average Rescue Time (ART), Average Imposed Waiting Time (AIWT) | - Strengths: The proposed SD-TLP mechanism achieved the best results compared against the baselines. - Limitations: The proposed approach is limited to a single emergency vehicle. |
| [19] | M. Zlatkovic et al., 2023 | To develop a connected vehicle-based algorithm that preempts traffic signals for emergency vehicles to reduce their delay. | VANET, Intelligent Transportation System | VISSIM | No pre-emption method | Vehicles (both emergency and non-emergency) delay, Travel Speed, Number of Stops | - Strengths: the proposed algorithm can significantly reduce emergency vehicle delays with minimal impact on regular vehicle traffic. - Limitation: Limited to a single vehicle. |
| [21] | A. Rego et al., 2018 | To propose a Software Defined Network (SDN) and Internet of Things (IOT) based architecture along with an algorithm for emergency traffic management in smart cities. | IOT, SDN, Smart cities | Mininet, VLC | Emergency scenario without the proposed algorithm. | Network delay (which is an abstraction of vehicle delay), communication energy consumption of IOT nodes. | - Strengths: the proposed architecture is scalable and potentially applicable to different smart cities. - Limitations: traffic is abstract. |
| [23] | P. Bairi et al., 2025 | EmV prioritization and traffic flow optimization | Traffic signal control suing VANET and V2I communication. | SUMO and OMNeT++ | Fixed-Time Control System (FTCM), Flexible Signal Preemption Method (FSPM), and Intrusive Signal Preemption Method (ISPM). | EmV travel time, average vehicle delay, and signal preemption success rate. | - Strength: prioritizes EmV in real-time - Limitation: depends on VANET communication. |
| [24] | Y. Chen et al., 2024 | EV traffic signal preemption considering queue spillbacks and non-priority traffic | Urban traffic management and emergency response | SUMO | Normal operation (no preemption) and an existing emergency preemption strategy [25]. | EV travel time, non-priority vehicle delay, and signal recovery time. | -Strength: handles queue spillback and reduces delays for non-priority vehicles. -Limitation: performance relies on real-time traffic and queue data. |

Table 2: Summary of Traffic Signal Control for Single Emergency Vehicle (continued)

| Ref. | Citation | Focus | Domain | Tool/ Simulator | Baselines | Performance measures | Notes (strength/limitation) |
|---|---|---|---|---|---|---|---|
| [26] | K.Shaaban et al., 2019 | A joint strategy for optimal path selection and EV preemption | Emergency vehicle routing, signal preemption | VISSIM | Normal traffic, with preemption | Travel Time, Intersection delay, Number of Stops | The approach performs joint optimization of path selection and emergency vehicle preemption using Dijkstra's algorithm with link cost, however, it does not account for real-time traffic dynamics. |
| [27] | L.Zhong et al., 2022 | Novel real-time traffic signal control strategy combining on-demand signal timing, emergency vehicle preemption, and a recovery cycle | Traffic Signal Control, Intelligent Transportation Systems | SUMO | FTCM (Fixed-Time Control Method), FSPM (Min's Flexible Signal Preemption Method), ISPM (Qin's Intrusive Signal Preemption Method) | Travel time, average travel time reduction | When compared with the baselines it had better travel time reduction results |
| [30] | P. Rosayyan et al., 2023 | Optimal control strategy for EV preemption using edge computing and IoT sensors | Smart cities, edge computing, IoT | Dash Board Unit and an edge server | Existing centralized EVP system | waiting time of normal vehicles | waiting time of normal vehicles on the other road are reduced by 73.23% in the proposed system when compared to the existing system |
| [31] | A.A.Salih et al., 2019 | Dynamic Preemption Logic to minimize Emergency Vehicle (EV) delay while stabilizing general traffic flow at signalized intersections. | Intelligent Transportation Systems (ITS), Smart City Infrastructure | VISSIM, using COM interface for external algorithm integration. | Static Preemption (Fixed triggers), Normal Signal Control (No priority). | EV Travel Time reduction, Network Delay (Impact on non-EV traffic), Queue Dissipation Rate | -Strengths: It uses early queue clearing to empty the path before the vehicle arrives, ensuring zero-stop travel. -Limitations: It lacks conflict logic, meaning it cannot handle two emergency vehicles coming from different directions at once |
| [28] | W. Min et al., 2020 | On-demand green wave combining reliable path-searching with elastic signal preemption | Time-Varying Networks, Reliable Routing, Signal Preemption | Real-world deployment using AutoNavi traffic data and regular GPS devices | Traditional shortest path methods, Fixed signal preemption | Response time, Overall traffic flow impact | Results demonstrated that the elastic signal preemption strategy reduced EV response time by about 30% |

## 5.2 Traffic Signal Control for Multiple Emergency Vehicles

Limited studies exist that deploy traffic light preemption mechanisms for scenarios involving multiple emergency vehicles. In this subsection, a few studies that address this problem using different mechanisms are reviewed. A summary of these works is provided in Table 3.

Younes and Boukerche [2] proposed a dynamic traffic light scheduling algorithm that adapts traffic light signal phases based on real-time traffic flow and the presence of one or more emergency vehicles. Furthermore, the algorithm is enhanced to handle situations where multiple emergency vehicles may arrive at the same intersection simultaneously. In such cases, priority is assigned by emergency vehicle type (ambulance, then fire truck, then police), and when emergency vehicles have the same priority type, priority is given to the one with the shortest Estimated Arrival Time (EAT) to the intersection. The authors evaluated the approach using SUMO [17] and NS-2 [32] on a single four-way intersection. Compared with ITLC [33] and OAF [34], the proposed algorithm increases the waiting delay of normal vehicles by 20% relative to ITLC, while reducing normal vehicle delay by approximately 10% compared to OAF. For emergency vehicles, the proposed method eliminates waiting delay at the intersection for a single emergency vehicle and, in scenarios involving multiple emergency vehicles, reduces emergency vehicle delays by 50% compared to ITLC and 60% compared to OAF. Overall, the proposed approach significantly improves scenarios that include multiple emergency vehicles, indicating a trade-off between emergency response priority and overall traffic efficiency.

Similarly, Humagain and Sinha [3] proposed an emergency vehicle preemption system that uses VANETs to gather real-time information about each emergency vehicle location, speed, route, and urgency. The proposed approach assigns different priorities to multiple emergency vehicles based on the severity of the emergency. Furthermore, the system calculates exactly how long traffic signals need to stay green at each intersection which will allow emergency vehicles to move efficiently while also at the same time minimizing delays for other vehicles. Simulations using SUMO [17], OMNeT++ [16], and Veins [18] show that this approach helps almost all emergency vehicles meet their target response times while also reducing waiting times and congestion for normal traffic.

Wang et al. [35] proposed a degree-of-priority-based control strategy for multiple emergency vehicles at intersections using a multilayer fuzzy model to assign priority levels. The model computes two main inputs: preemption demand intensity, based on estimated economic loss, number of injuries, and urgency, and preemption influence intensity, based on road type and traffic saturation. The strategy then adjusts traffic signal timings by modifying the offsets between intersections to create a continuous green-wave for emergency vehicles, and includes lane or road clearance for highest-priority cases. Optimal paths are selected using Dijkstra's algorithm [15]. The approach was evaluated on a real urban network in Xicheng District, Beijing, using VISSIM [20] and cellular automata-based simulations. Results showed substantial reductions in emergency vehicle delays while minimizing disruption to normal traffic. However, the study relies on simulation and was tested only in a single urban area.

Lin et al. [5] introduced a sophisticated cooperative priority model for multiple emergency vehicles (EMVs) within a connected vehicle (CV) environment, moving beyond traditional signal preemption to integrated trajectory optimization. The researchers formulated the problem as an Integer Linear Programming (ILP) model that simultaneously optimizes the speed, acceleration, and lane-changing maneuvers of both EMVs and surrounding ordinary vehicles (OVs). To address the computational complexity, they compared their method against several baselines, including First-Come-First-Serve (FCFS) and Conventional Preemption. The results, validated via SUMO [17], demonstrated that while traditional preemption clears signals, their cooperative model clears the actual physical path (lane pre-clearing), significantly reducing EMV response times and minimizing the disturbance on normal traffic. However, the study notes that the system's success relies on high-speed V2X communication and substantial computational resources for real-time decision-making.

He et al. [4] proposed a multimodal traffic signal control that integrates transit and pedestrian priority with signal actuation and arterial coordination. The study addresses the challenge of serving simultaneous priority requests from different modes, such as emergency vehicles, buses, and pedestrians. The control strategy is formulated as a mixed integer linear programming (MILP) model and is evaluated using VISSIM [20], and the priority requests were generated through an assumed V2I communication. Furthermore, the performance was compared with actuated coordinated control and coordinated transit signal priority using measures such as average delay for buses, passenger vehicles, and pedestrians, as well as travel time, number of stops, queue length, throughput, and priority service reliability. The simulation results indicate that the proposed method reduces delays across all modes. Key strengths of the study include its ability to serve multiple requests and the efficient use of actuation to reduce the unused green time. However, the method depends on V2I equipped users which may limit its scalability.

Table 3: Summary of Traffic Signal Control for Multiple Emergency Vehicles

| Ref. | Citation | Focus | Domain | Tool/ Simulator | Baselines | Performance measures | Notes (strength/limitation) |
|---|---|---|---|---|---|---|---|
| [2] | M. B. Younes and A. Boukerche, 2014 | Dynamic traffic light scheduling algorithm with single and multiple emergency vehicle priority | VANET, Intelligent Transportation System | SUMO, NS-2 | ITLC algorithm, OAF algorithm | Average waiting delay time for regular (non-emergency) vehicles, Throughput, Average waiting delay time for emergency vehicles | -Strengths: Handles multiple emergency vehicles.<br>-The proposed method indicate a trade-off between emergency response priority and overall traffic efficiency. |
| [3] | S. Humagain and R. Sinha, 2020 | Emergency vehicle pre-emption for multiple vehicles with different criticality levels. | Traffic signal control and emergency vehicle routing. | SUMO, OMNeT++, and Viens | No preemption and absolute preemption. | EV meeting target time, average waiting time, and intersection queue length | -Strengths: maps priority levels to criticality levels to manage multiple EVs simultaneously.<br>-Limitations: Relies on V2I equipped vehicles and pedestrians. |
| [35] | J. Wang et al., 2013 | Degree-of-priority control strategy for coordinating multiple emergency vehicles using fuzzy-based prioritization and green-wave signal control | Emergency Vehicle Preemption, ITS, Urban Traffic Control | VISSIM + Cellular Automata–based simulation | Conventional local preemption method | Emergency vehicle delay, impact on normal traffic flow, network performance | - Strengths: Multilayer fuzzy model uses offset adjustment to create green-wave and integrates lane/road clearance.<br>-Limitations: Simulation-based validation only and is evaluated in a single city network scenario. |
| [5] | P. Lin et al., 2024 | Simultaneous control of signal timing and individual vehicle maneuvers (speed and lane changes) to create an automated ”green corridor” | Connected and Automated Vehicles (CAVs), VANETs | SUMO | First-Come-First -Serve (FCFS), Conventional Preemption. | EMV Response Time, Average System Delay,. Throughput. | - Strengths: Effectively resolves priority conflicts when multiple emergency vehicles arrive at the intersection simultaneously.<br>-Limitations: High complexity and makes real-time scaling for large urban networks challenging. |
| [4] | Q. He et al., 2014 | Integration of real-time priority handling in traffic signal control to manage simultaneous requests | Multimodal traffic signal control with transit and pedestrian priority | VISSIM | ASC Coordination, ASC-TSP Coordination. | Average delay, travel time, number of stops, queue length, intersection throughput, and priority requests service rates. | -Strengths: Integrates multimodal priority, actuation, and coordination in a single framework and demonstrates robust performance under high demand. |

### 5.3 Base Paper Summary

In this section, we summarize the work of Bagheri. et al [1], which is the foundation for our work. The summary of the paper is shown in Table 4.

Table 4: Base Paper Summary

| **Field** | **Entry** |
|---|---|
| Base paper full citation + link (DOI/URL) | N. Bagheri, S. Yousefi, and G. Ferrari, "Software-defined traffic light preemption for faster emergency medical service response in smart cities," *Accident Analysis & Prevention*, vol. 196, 2024, Art. no. 107425.<br>DOI: `https://doi.org/10.1016/j.aap.2023.107425` |
| What is reproduced (figure/table/metric) | Reproduced Algorithm 3 (SD-TLP Algorithm) from the base paper. Also, the same performance metrics from the base paper were used:<br>• Average Rescue Time (ART)<br>• Average Imposed Waiting Time (AIWT)<br>• Average Waiting Time (AWT) |
| Code/artifacts used (link) or how implemented | Implemented using OMNeT++ version 6.3.0. |
| Key assumptions adopted from the paper | • Centralized SDN controller with global traffic view.<br>• Emergency vehicle speed and traffic parameters are predefined.<br>• Emergency vehicles have a 100% penetration rate. |

### 5.4 Discussion

Based on the previously reviewed studies [1,19,23,24], traffic light preemption and prioritization have been widely explored for improving emergency vehicles' response time, allowing them to reach the accident destination quickly and efficiently while trying to minimize normal traffic delay. This shows the importance of granting priority to emergency vehicles, since delays caused by traffic and red lights may affect survival rates, fire spread, and overall emergency response.

Different approaches have been investigated for emergency vehicle traffic light preemption. Some of these approaches include using a centralized controller with a global view of traffic conditions that follows a preemption algorithm, as done by Bagheri et al. [1] and Rego et al. [21], a traffic signal preemption strategy that highly depends on queue information [24,31], or a preemption strategy that triggers when the emergency vehicles enters a fixed detection distance [19]. In addition, although SDN-based traffic light preemption approaches exist, such as the one proposed by Rego et al. [21], the SD-TLP mechanism by Bagheri et al. [1] differs in terms of how preemption requests are handled, and further incorporates optimal route selection using Dijkstra's algorithm [15] and lane changing mechanisms.

Despite the effectiveness of these approaches, they focused on scenarios involving a single emergency vehicle and did not explicitly address scenarios where multiple emergency vehicles, whether of the same or different priority levels, simultaneously request preemption. As a result, this may lead to conflicts and significant delays in emergency response. To address these limitations, some studies [2, 3, 35] have extended traffic signal preemption strategies to explicitly consider scenarios involving multiple emergency vehicles. These works can be categorized into three paradigms based on how they resolve conflicts between simultaneous preemption requests, summarized in Table 5. The first paradigm consists of decentralized, local intersection rule-based approaches, where conflict resolution decisions are made independently at each intersection, such as vehicle type priority followed by estimated arrival time, as done by Younes and Boukerche [2], or severity-based priority, as done by Humagain and Sinha [3]. The second paradigm consists of centralized optimization approaches, which resolve conflicts through complex signal control strategies such as multilayer fuzzy modeling [35] and mixed-integer linear programming (MILP) [4, 5]. While these approaches achieve reductions in emergency vehicle delays, they have a high computational cost [5]. The third paradigm, and the one most relevant to our work, consists of centralized rule-based approaches, where a central controller with a global view applies simple priority rules to resolve concurrent preemption requests.

Table 5: Three paradigms for multi-emergency-vehicle traffic-light preemption. The bottom-right cell — the explicit transfer mechanism and its empirical characterization — is the gap this work addresses.

| | **Decentralized rule-based** [2, 3] | **Centralized optimization** [4, 5, 35] | **Centralized rule-based** **(this work)** |
|---|---|---|---|
| **Where decisions are made** | Each intersection, independently | Central controller with global view | Central SDN controller with global view |
| **Decision mechanism** | Vehicle type, then EAT | MILP or fuzzy model over many requests | Vehicle type, then EAT, plus dynamic transfer |
| **Computational cost at runtime** | Low | High | Low |
| **Mid-cycle transfer when a higher-priority EV arrives** | Not supported | Implicit in re-optimization | Explicit, characterized in this work |
| **Evaluated under multi-EV conditions** | Yes | Yes | Not in the published literature, to our knowledge |

To our knowledge, the centralized rule-based combination for traffic light preemption has not been evaluated under multiple emergency vehicle conditions in the published literature. In particular, limited work has investigated centralized rule-based mechanisms that dynamically re-evaluate and transfer active traffic light preemption sessions when a higher priority emergency vehicle becomes eligible.

To address this gap, this paper investigates centralized rule-based multi-emergency vehicle traffic light preemption within the SD-TLP framework [1], introducing and empirically characterizing a dynamic preemption transfer mechanism that re-evaluates active preemption sessions and transfers priority to a higher priority emergency vehicle when it becomes eligible, while identifying the traffic conditions in which this approach is and is not beneficial.

Moreover, from the reviewed work, multiple studies [1, 19, 21, 24] have used the No Preemption mechanism as a baseline, indicating that it is an accepted approach for evaluating the effectiveness of traffic light preemption methods. In addition, [5] used First Come, First Serve (FCFS) to compare their proposed approach in scenarios involving multiple emergency vehicles. Since the SD-TLP mechanism proposed by Bagheri et al. [1] does not support more than one emergency vehicle, we incorporated two alternative conflict resolution policies within the SD-TLP framework to serve as internal baselines, SD-TLP with FCFS and SD-TLP Closest. The SD-TLP with FCFS baseline prioritizes the emergency vehicle that first requests preemption, while the SD-TLP Closest prioritizes the emergency vehicle with the shortest remaining geographic distance to the accident location. The SD-TLP Closest baseline is based on the intuition that the emergency vehicle closer to the accident location has less remaining distance to travel, and therefore may reach the destination faster when given preemption priority. Comparison against decentralized local intersection approaches, such as Younes and Boukerche [2] is left for future work.

# 6. Proposed Solution

In this work, we investigated a centralized rule-based traffic light preemption mechanism within the Software-Defined Traffic Light Preemption (SD-TLP) framework proposed by Bagheri et al [1]. Our proposed solution preserves the original three-part SD-TLP structure while using a rule-based decision mechanism for resolving conflicts when multiple emergency vehicles request traffic light preemption simultaneously. In addition, we include a dynamic preemption transfer mechanism, where the centralized controller dynamically transfers priority to a higher-priority emergency vehicle when it enters its detection distance (DD) and becomes eligible for preemption.

## 6.1 System Overview

Efficient traffic signal preemption is important for reducing the rescue time of emergency vehicles while minimizing delays for regular traffic. In addition, real-world scenarios often involve multiple emergency vehicles with different priority levels simultaneously requesting traffic light preemption. Therefore, in this paper, we address these challenges by considering the cases that involve one or more emergency vehicles with either the same or different priority levels.

Our work relies on a software-defined controller similar to the one in [1], that maintains a global view of the city traffic and is responsible for executing the traffic control algorithm, as well as managing traffic signal operations, including granting preemption to emergency vehicles. We assume that the accident has already been reported at a fixed location, the emergency vehicle is predetermined and has been dispatched, and that its route is fixed.

The data flow in our approach is shown in Figure 1 and works as follows. We first assume that there are four roadside units (RSUs), one at each intersection of our map. These RSUs act as data collectors that receive information from emergency vehicles and forward it to the SDN-based centralized controller, enabling Vehicle-to-Infrastructure (V2I) communication. Typically, an RSU can be IEEE 802.11p or 5G enabled [36]. In our methodology, we assume that the emergency vehicle will provide information to the RSUs, including their speed, location, accident position, type (ambulance, police, or fire truck ), and the direction or approach from which they are coming from.

Additionally, the traffic information, such as the number of vehicles on the road segment, is collected by the traffic light controller. This information is also passed to the centralized controller. The centralized controller then uses the received information to execute a traffic light preemption algorithm and sends the preemption decisions to the traffic lights through the southbound interface, as illustrated in Figure 1.

It is important to note that in this work, we assume perfect and lossless knowledge of vehicle counts and emergency vehicle positions. Specifically, we assume that the number of vehicles on each road is always accurately known by the centralized controller, and that emergency vehicles continuously report their speed, location, type, and the approach they are coming from with no latency and loss. This assumption is further discussed in Section 9.

In our system, when there is a single emergency vehicle, the algorithm follows a similar approach to the one proposed in [1]. However, when multiple emergency vehicles are present, the centralized controller resolves the concurrent preemption requests using a rule-based policy and a dynamic preemption transfer mechanism. Based on this, below we describe the Software-Defined Traffic Light Preemption (SD-TLP) algorithm proposed by Bagheri et al. [1], followed by the proposed multi-emergency vehicle SD-TLP algorithm to support multiple emergency vehicles.

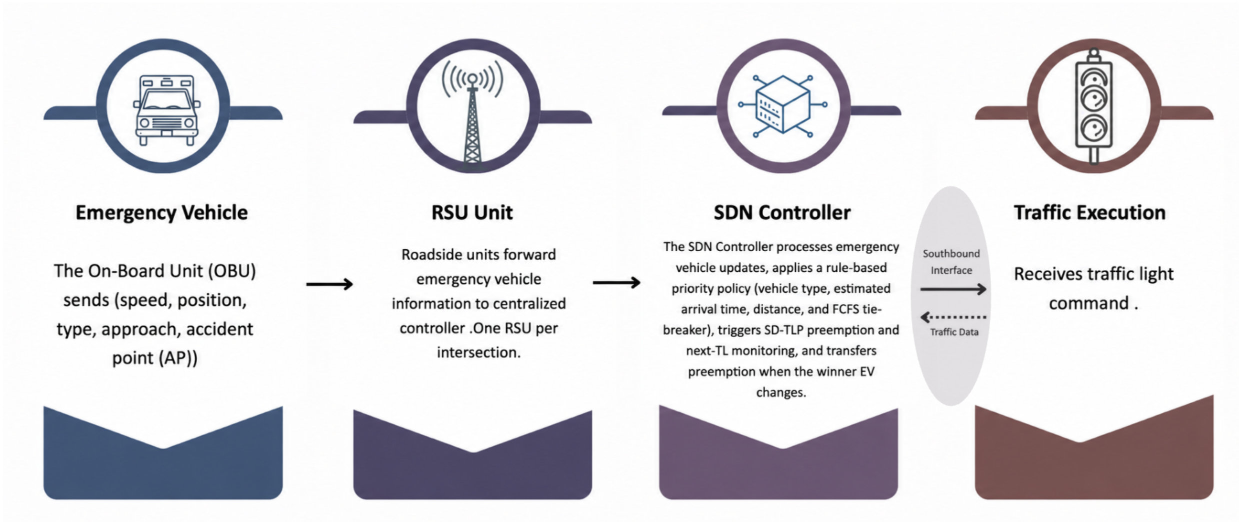



Figure 1: High-Level System Overview of the Proposed Framework Under Idealized Sensing and Communication Assumptions.

### 6.2 Software-Defined Traffic Light Preemption (SD-TLP) Mechanism

To better understand our proposed approach, we first provide a detailed explanation of the Software-Defined Traffic Light Preemption (SD-TLP) mechanism proposed by Bagheri et al. [1]. The SD-TLP algorithm consists of three main parts. The first part of the algorithm determines the appropriate time and location for the preemption operation to begin. To achieve this, the emergency vehicle (EV) continuously sends its location to the roadside units (RSU) until it reaches the accident scene, as shown in Lines 8-9 of Algorithm 1. When the EV enters a new road segment, it is detected by the controller. At that point, two values are calculated: the estimated time (ET) needed to clear the queue of vehicles in the current road segment and the detection distance (DD).

The ET represents the time required to clear the vehicles in front of the EV. It is calculated using Equation (1), where $C$ is the number of vehicles on the road segment, $L$ is the average vehicle length, $MG$ is the minimum gap between vehicles, $S$ is the average speed of normal vehicles, and $r$ is the traffic signal switching delay.

$$ET = \frac{C \cdot L + (C-1) \cdot MG}{S} + r \tag{1}$$

The detection distance (DD) is the distance from the EV to the traffic light at which the preemption operation should begin. It is calculated by multiplying the estimated time obtained from Equation (1) by the EV speed (EVS), as shown in Equation (2).

$$DD = ET \cdot EVS \tag{2}$$

If the EV is within the detection distance, the traffic light is adjusted based on its current phase. If the current phase is red, then the red phase will be shortened by first switching the light to yellow for three seconds, followed by an all-red phase for three seconds to ensure intersection safety, after which the EV's approach is switched to green. If the current phase is yellow, it changes immediately to green, and if the phase is already green, the green time is extended.

The second part of the algorithm aims to maintain a smooth flow for the EV by continuously monitoring the next two traffic lights along the rescue route. If the traffic density in either of the road segments exceeds a predefined threshold, then the traffic light of the corresponding road segment is switched to green, as indicated in lines 30-42 of Algorithm 1 .

The third part of the algorithm deals with the traffic light recovery mechanism. After the EV passes an intersection, the distance between the current traffic light and the previous one is calculated. If this distance exceeds a predefined threshold, only the current traffic light is switched to red. However, if the distance is below the threshold, both traffic lights are switched to red to prevent vehicles from the previous intersection from immediately entering the next one.

**Algorithm 1** Software-defined Traffic Light Preemption (SD-TLP)

```
Input: EV location, Traffic light locations on the rescue route, AP, C, L, MG, S, EVS
Action: Determine appropriate time and location to start preemption and change traffic
    lights to green
Symbols: EV: Emergency Vehicle ID, AP: Accident Point, TL: Traffic Light ID,
Dtls: Distance between two consecutive traffic lights,
Dthreshold: Threshold distance between two consecutive intersections,
DEMV: Distance of EV to traffic light, TD: Traffic Density.
function SDX-BASED PREEMPTION(AP)
    while EV has not reached AP do
        EV sends road ID
        if it is a new road segment then
            Compute ET using Equation (1)
            Compute DD using Equation (2)
            for previous traffic lights do          ▷ traffic lights recovery mechanism
                if Dtls < Dthreshold then
                    Set the second TL phase as the first TL phase
                end if
            end for
        end if
        if D_EMV ≤ DD then                          ▷ EV within detection distance
            if TLphase = Red then
                Reduce red phase duration
            end if
            if TLphase = Yellow then
                Change immediately to green
            end if
            if TLphase = Green then
                Extend green phase
            end if
        end if
        for next two lights do
            if TD > threshold then                  ▷ high traffic density
                if TLphase = Red then
                    Reduce red phase duration
                end if
                if TLphase = Yellow then
                    Change immediately to green
                end if
                if TLphase = Green then
                    Extend green phase
                end if
            end if
        end for
    end while
end function
```

### 6.3 Centralized Rule-Based Multi-Emergency-Vehicle SD-TLP Mechanism

In this work, we investigated a centralized rule-based traffic light preemption mechanism within the Software-Defined Traffic Light Preemption (SD-TLP) framework proposed by Bagheri et al. [1], including a dynamic preemption transfer mechanism in which the centralized controller dynamically transfers traffic light preemption priority to a higher-priority emergency vehicle when it enters the detection distance (DD) and becomes eligible for preemption. The proposed mechanism preserves the same three main parts of the SD-TLP algorithm.

The first part focuses on calculating the estimated time (ET) needed to clear the queue of vehicles in front of the emergency vehicle and the detection distance (DD) at which the emergency vehicle becomes eligible for preemption. The second part monitors the state of the next two traffic lights along the rescue route and turns any of them to green if the traffic load of the corresponding road segment is greater than 70%. The third part focuses on a post-preemption strategy that changes the appropriate traffic lights to red in order to reduce the negative effects on normal traffic. Algorithm 2 shows the SD-TLP algorithm that supports multiple emergency vehicles.

When multiple emergency vehicles approaching from different directions become eligible for preemption upon entering the detection distance (DD), the centralized controller selects the emergency vehicle whose approach will be granted traffic light preemption based on the following priority rules:

1. **Emergency vehicle type:** Priority is first determined based on the type of emergency vehicle, where ambulances are given the highest priority, followed by fire trucks, then police vehicles, as shown in lines 10-11 of Algorithm 3. This order follows Younes and Boukerche [2], where they stated that this ordering is based on the global driving rules.

2. **Estimated arrival time (EAT):** If the emergency vehicles are of the same type, the emergency vehicle with the shortest estimated arrival time (EAT) to the intersection is given higher priority, as shown in lines 12-16 of Algorithm 3, following [2]. The estimated arrival time (EAT) to the intersection is computed using Equation (3), where $\text{DEMV}_i$ represents the distance between emergency vehicle $i$ and the signalized intersection, and $S\text{ev}$ represents the emergency vehicle's speed.

$$\mathrm{EAT}_i = \frac{\mathrm{DEMV}_i}{S\mathrm{ev}} \tag{3}$$

3. **Distance to accident location:** In cases where multiple emergency vehicles have equal EAT, priority is given to the emergency vehicle that is closer to the accident location, as shown in lines 20-22 of Algorithm 3. This rule is motivated by the intuition that the emergency vehicle with the shortest remaining geographical distance to the accident position has less distance to cover and may therefore reach the accident position faster when given preemption priority.

4. **First Come, First Serve (FCFS):** If the emergency vehicles are still equal after applying the previous rules, the algorithm follows a first come, first serve (FCFS) approach, where the emergency vehicle that entered detection distance (DD) first is given higher priority, as shown in lines 18-20 of Algorithm 3.

Moreover, in scenarios where an emergency vehicle with a higher priority than the one currently being served becomes eligible, the centralized controller performs the following steps:

1. The traffic light preemption for the emergency vehicle with the lower priority ends.

2. Traffic light preemption for the emergency vehicle with the higher priority begins.

This process represents the proposed dynamic transfer mechanism that is evaluated in this work. Figure 2 illustrates the dynamic transfer mechanism, showing how transfer occurs when a higher priority emergency vehicle enters its detection distance (DD).

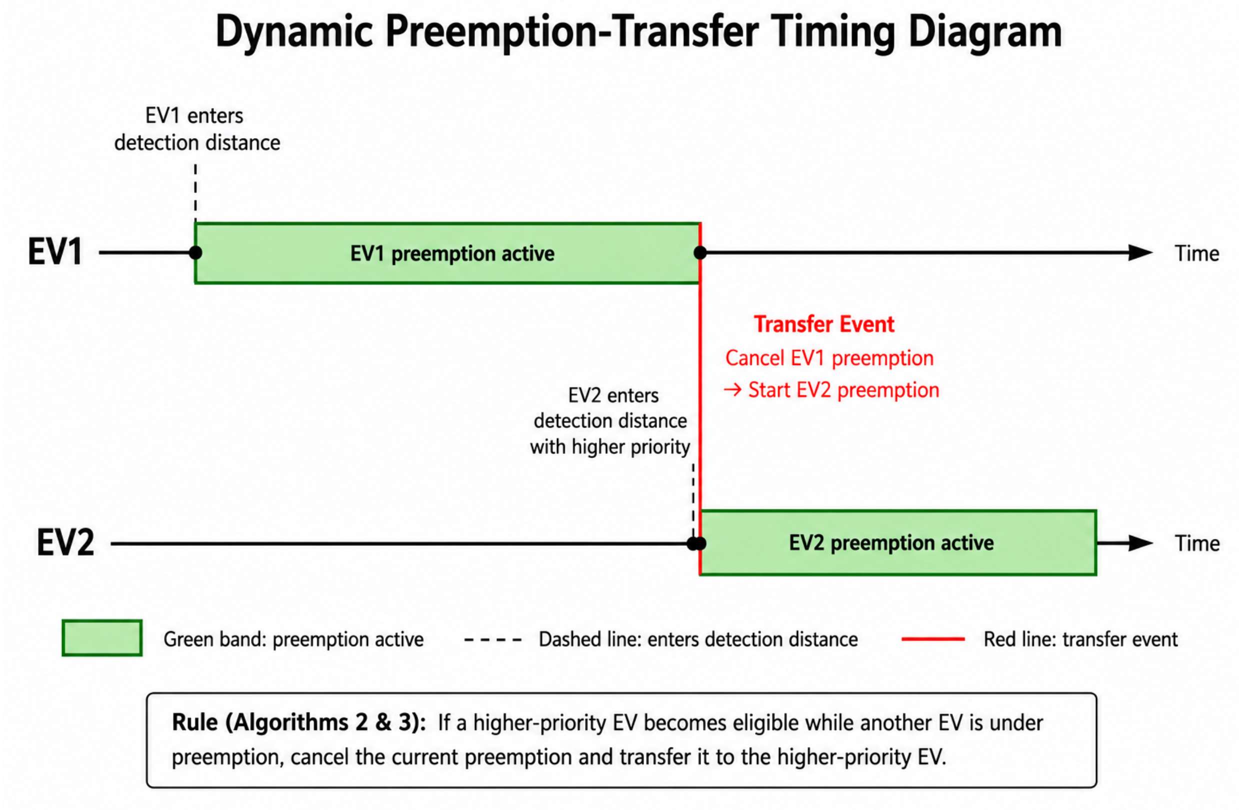



Figure 2: Dynamic preemption-transfer timing diagram.

The parameters used to implement this method are presented in Section 7.5. The proposed vehicle type prioritization strategy is chosen to ensure that the most critical emergency vehicles are selected first, where ambulances are considered the most critical in this work, followed by fire trucks, then police vehicles. This is based on the ethical consideration study presented in Section 8.6. In addition, the use of the estimated arrival time (EAT) helps select the vehicle that can reach the intersection faster, reducing waiting time. The distance condition helps resolve ties by selecting the emergency vehicle closer to the accident position. Lastly, the FCFS rule is used to ensure fairness when all conditions are equal. Overall, the proposed mechanism aims to reduce emergency vehicle rescue time while minimizing the impact of traffic light preemption on normal traffic.

**Algorithm 2** Multi-Emergency Vehicle Software-defined Traffic Light Preemption

**Input:** EV, priority, traffic light locations in the rescue route, DEMV, $S_{ev}$, $DistToAP$, $AP$, $C$, $L$, $MG$, $S$

**Action:** Select the appropriate EV to serve, and apply recovery and safe override.

**Symbols:** EVs: set of emergency vehicles, $AP$: Accident point, $C$: Number of vehicles present on the road segment, $L$: Average vehicle length, $MG$: Minimum gap between vehicles, TL: Traffic light id, $DEMV$: Distance of EV to current TL, $S_{ev}$: EV speed, $S$: Speed of normal vehicles, $TD$: Traffic density of a road segment, $Dtls$: Distance between consecutive traffic lights, $Dthreshold$: Threshold distance between two consecutive TLs, $EAT$: Estimated Arrival Time to intersection, $ET$: Estimated time, $DistToAP$: EV distance to accident position, $DD$: Detection distance

**function** SDX-based Preemption Multi (AP):
  **while** EVs did not reach AP **do**
    EVs sends status update to the controller (including road id)
    **if** it is a new road segment **then**
      Compute $ET$ using equation (1)
      Compute $DD$ using equation (2)
      **for** previous traffic lights **do** ▷ part 3 of the SD-TLP algorithm
        **if** $Dtls < Dthreshold$ **then**
          Set the second TL phase as the first TL phase
        **end if**
      **end for**
    **end if**
    SelectWinnerEV() ▷ a function that selects the 'winner' EV based on priority rules

*// Dynamic preemption transfer — proposed in this work*
    **if** WinnerEV changed **then**
      End current preemption
      Start new preemption
    **end if**

      **if** $DEMV_{Winner} \leq DD$ **then** ▷ part 1 of the SD-TLP algorithm
        **if** $TLphase = Red$ **then**
          Reduce red phase duration
        **end if**
        **if** $TLphase = yellow$ **then**
          Change immediately to green
        **end if**
        **if** $TLphase = Green$ **then**
          Extend green phase
        **end if**
      **end if**
      **for** next two lights **do** ▷ part 2 of the SD-TLP algorithm
        **if** $TD > TDthreshold$ **then**
          **if** $TLphase = Red$ **then**
            Reduce red phase duration
          **end if**
          **if** $TLphase = yellow$ **then**
            Change immediately to green
          **end if**
          **if** $TLphase = Green$ **then**
            Extend green phase
          **end if**
        **end if**
      **end for**
    **end while**
  **end function**

**Algorithm 3** Winner EV Selection Function

**Action:** Function that selects the winner EV based on the priority rules

**Symbols:** $DEMV$: Distance of EV to current TL, $S_{ev}$: EV speed, $EAT$: Estimated Arrival Time to intersection, $DistToAP$: EV distance to accident position, $DD$: Detection distance, $TDD$: Time at which the EV enters the detection distance.

**function** SELECTWINNEREV
    $WinnerEV \leftarrow Null$
    **for** each $EV_i$ **do**
        **if** $DEMV_i \leq DD$ **then**
            **if** $WinnerEV = NULL$ **then**
                $WinnerEV \leftarrow EV_i$
            **end if**
            **if** $TypePriority_i > TypePriority_{Winner}$ **then**
                $WinnerEV \leftarrow EV_i$
            **else if** $TypePriority_i = TypePriority_{Winner}$ **then**
                Compute $EAT_i = \frac{DEMV_i}{S_{ev}}$
                Compute $EAT_{Winner} = \frac{DEMV_{Winner}}{S_{ev}}$
                **if** $EAT_i < EAT_{Winner}$ **then**
                    $WinnerEV \leftarrow EV_i$
                **else if** $EAT_i = EAT_{Winner}$ **then**
                    **if** $DistToAP_i < DistToAP_{Winner}$ **then**
                        $WinnerEV \leftarrow EV_i$
                    **else if** $DistToAP_i = DistToAP_{Winner}$ **then**
                        **if** $TDD_i < TDD_{Winner}$ **then**
                            $WinnerEV \leftarrow EV_i$
                        **end if**
                    **end if**
                **end if**
            **end if**
        **end if**
    **end for**
**end function**

# 7. Experimental Setup and Methodology

This section describes the experimental setup used to evaluate the proposed multi-emergency vehicle SD-TLP system. It outlines the simulation environment, the parameters used in the experiments, the evaluated scenarios, the considered baselines, and the performance metrics used to assess the system.

## 7.1 Tools and Environment

For the simulation, OMNeT++ version 6.3.0 [16] was used as the simulator to model city traffic and traffic signal control. OMNeT++ is a C++ simulation library and framework used for building network simulations. The simulation network contains one controller, four intersections, four RSUs, and a number of emergency vehicles (EVs) depending on the scenario. The implemented simulation is a simplified and abstract model of the study done by Bagheri et al. [1], which used OMNeT++, SUMO [17], and Veins [18] to simulate realistic traffic movement, while this work uses OMNeT++ only. As a result, vehicle movement and traffic conditions are represented in a simplified way. Since there is no publicly available code for the SD-TLP system done by Bagehri et al. [1], some simulation parameters are based on assumptions derived from the available description in their paper and the simulation map used in this study.

## 7.2 Scenarios

The simulation map consists of four intersections connected by road segments extending in opposite directions. Following [37], each road segment has a length of 300 m. Additionally, Road Side Units (RSUs) are used to receive traffic information from vehicles and transmit it to the centralized controller [1]. One RSU is deployed at each intersection in our map, as suggested in [38]. A single SDN controller manages traffic signal operations across the simulation map. The map used in the experiments is shown in Figure 3. The simulation will be tested in different scenarios under different traffic conditions to evaluate and assess the performance of our proposed solution.

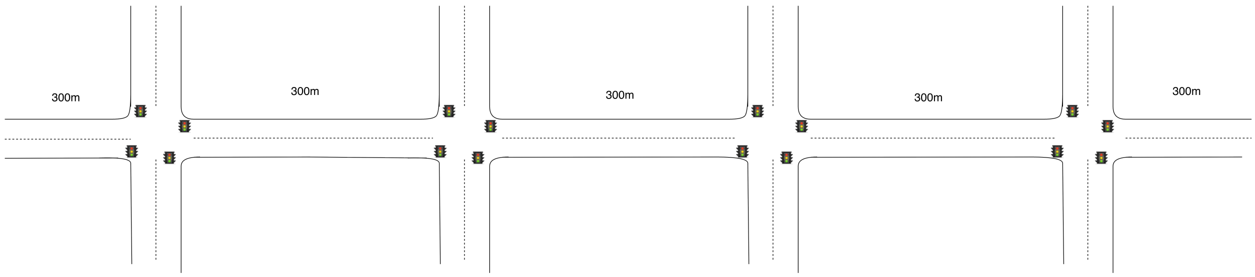



Figure 3: The proposed simulation map used in the experiments.

**Scenario 1: One Emergency Vehicle.**
In this scenario, only one emergency vehicle (EV1) is present. EV1 starts at a random position on the west approach, between 50 m and 300 m from Intersection 1, and moves east. Its accident location is 300 m east of Intersection 1. Since only one emergency vehicle exists in the network, no conflict occurs at the intersection. This scenario is used to reproduce the original SD-TLP algorithm proposed in [1] under different traffic arrival rates.

**Scenario 2: Two Emergency Vehicles with Different Priority.**
In this scenario, two emergency vehicles approach Intersection 1 from opposite directions. EV1 starts at a random position on the west approach, between 50 m and 300 m from Intersection 1, and moves east. EV2 starts at a random position on the east approach, between 50 m and 300 m from Intersection 1, and moves west. The accident location of EV1 is 300 m east of Intersection 1, while the accident location of EV2 is 300 m west of Intersection 1. EV1 has higher priority than EV2. This scenario is used to evaluate how the system handles a conflict between two emergency vehicles with different priority levels.

**Scenario 3: Two Emergency Vehicles with the Same**
In this scenario, two emergency vehicles (EV1 and EV2) with the same priority approach Intersection 1 from opposite directions. EV1 starts at a random position on the west approach, between 50 m and 300 m from Intersection 1, and moves east. EV2 starts at a random position on the east approach, between 50 m and 300 m from Intersection 1, and moves west. The accident location of EV1 is 300 m east of Intersection 1, while the accident location of EV2 is 300 m west of Intersection 1. EV2 starts 6 seconds after EV1. Since both vehicles have the same priority level, the conflict is resolved using the Estimated Arrival Time (EAT). This scenario evaluates whether the system correctly handles conflicts when

emergency vehicles have equal priority.

**Scenario 4: Three Emergency Vehicles.**
In this scenario, three emergency vehicles (EV1, EV2, and EV3) approach Intersection 1 from different directions. EV1 starts at a random position on the west approach, between 50 m and 300 m from Intersection 1, and moves east. EV2 starts at a random position on the east approach, between 50 m and 300 m from Intersection 1, and moves west. EV3 starts at a random position on the north approach, between 50 m and 250 m from Intersection 1, and moves south. The accident location of EV1 is 300 m east of Intersection 1, the accident location of EV2 is 300 m west of Intersection 1, and the accident location of EV3 is 200 m south of Intersection 1. EV1 and EV2 have the same priority, while EV3 has a higher priority. EV2 starts 3 seconds after EV1, and EV3 starts 6 seconds after EV1. This scenario evaluates the behavior of the system when multiple emergency vehicles approach the intersection and request preemption at similar times.

Scenario 1 was evaluated using the original SD-TLP algorithm and the No Preemption case as the baseline for reproducibility. The remaining scenarios were evaluated using four approaches: SD-TLP Closest, SD-TLP with First-Come-First-Serve (FCFS), the proposed SD-TLP multi-emergency vehicle algorithm, and the No Preemption case. In addition, each configuration was tested under three traffic levels: low, medium, and high traffic. Every configuration was executed using ten different random seeds. As a result, a total of 420 runs were conducted, and the final results were obtained by calculating the average of the ten runs.

### 7.3 Baselines

This section describes the baselines used for comparison and to evaluate the effectiveness of the proposed extension.

**Baseline 1: Software-Defined Traffic Light Preemption Mechanism (SD-TLP) with First Come, First Serve (FCFS)**

The first baseline is a modified version of the original Software-Defined Traffic Light Preemption (SD-TLP) algorithm proposed by Bagheri et al. [1], in which the First Come, First Served (FCFS) rule is applied. This version allows the system to handle situations where multiple emergency vehicles approach the same intersection, whereas the original SD-TLP does not include a mechanism for resolving simultaneous preemption requests from multiple emergency vehicles and would therefore lead to inconsistencies. In the modified version, when more than one emergency vehicle requests preemption, the vehicle that arrives first at the detection distance (DD) is granted traffic signal preemption. This baseline is only used for comparison within our centralized SD-TLP system, it is not a reproduction of the decentralized intersection-local approach proposed by [2].

**Baseline 2: Software-Defined Traffic Light Preemption Mechanism (SD-TLP) Closest**

The second baseline is another modified version of the original SD-TLP algorithm proposed by Bagheri et al. [1]. In this baseline, if more than one emergency vehicle satisfies the preemption eligibility condition at the same intersection, the controller selects the vehicle with the shortest remaining distance to its accident location. This baseline allows the system to handle multiple emergency vehicle requests using a distance-based selection rule. However, after an emergency vehicle is selected, the preemption continues until the current cycle is completed, and the controller does not perform mid-cycle switching or dynamic transfer if another emergency vehicle becomes eligible later.

**Baseline 3: No Preemption**

The third baseline represents the No Preemption case. In this scenario, emergency vehicles follow the normal traffic signals along their route without any priority. Traffic signals operate based on pre-programmed signal timings, representing standard traffic signal operation. By comparing our approach against this baseline, we evaluate the impact of traffic light preemption on emergency vehicle response and waiting times for normal vehicles.

### 7.4 Performance Metrics

The performance of the system is evaluated using several metrics adopted from [1]. These metrics measure the effect of the traffic signal control methods on both emergency vehicles and normal vehicles. In this study, the general performance metrics are used to evaluate rescue time, normal vehicle waiting time, and controller activity, while the transfer mechanism characterization metrics are used to analyze dynamic preemption transfer events.

### 7.4.1 General Performance Metrics

The general performance metrics are used to evaluate the overall performance of each traffic signal control method. These metrics measure the rescue time of emergency vehicles, the waiting time imposed on normal vehicles, and the number of control actions generated by the controller.

- **Average Rescue Time (ART):** The average time required for an emergency vehicle (EV) to travel from its starting position until it reaches the incident location. In the simulation, this value is measured as the time difference between the EV start time and the time it reaches its destination, as shown below:

$$ART = t_{\text{arrival}} - t_{\text{start}} \tag{4}$$

- **Average Imposed Waiting Time (AIWT):** The average waiting time experienced by normal vehicles on routes that intersect with the emergency vehicle's path. This metric shows how much delay is caused to other vehicles when priority is given to the emergency vehicle, as illustrated in (5).

$$AIWT = \frac{\text{Total waiting time of intersecting vehicles}}{\text{Number of intersecting vehicles}} \tag{5}$$

- **Average Waiting Time on Rescue Route (AWT):** The average waiting time experienced by normal vehicles traveling on the same route as the emergency vehicle toward the incident location, as mentioned in (6).

$$AWT = \frac{\text{Total waiting time of vehicles on the rescue route}}{\text{Number of vehicles on the rescue route}} \tag{6}$$

- **Controller Overhead:** The total number of additional actions that will be done by the SDN-based controller compared to normal signal traffic operation. In this study, it is measured as the number of preempt and recovery commands issued during each simulation run. This metric is important because a high number of control actions increases the workload on the controller. If too many preemption and recovery messages are generated, the controller may experience delays in processing or sending control commands. This may also increase communication delays or packet loss and limit the scalability of the system in real-world networks.

Since no explicit formula is provided in [1], AIWT and AWT are measured starting from the arrival of the first emergency vehicle until 5 seconds after the last emergency vehicle reaches its destination.

### 7.4.2 Transfer Mechanism Characterization Metrics

In addition to the main performance metrics, these metrics are used to characterize the dynamic preemption transfer mechanism. They measure how often transfers occur and how each transfer affects the higher-priority and displaced emergency vehicles. Transfer Frequency is reported per simulation run, while Rescue-Time Benefit, Rescue-Time Cost, and Net Rescue-Time Delta are calculated per transfer event by comparing the proposed approach with the two baseline approaches: SD-TLP with FCFS and SD-TLP Closest.

- **Transfer Frequency:** The average number of transfer events that occur per simulation run under each traffic condition, as shown in (7). It is used to evaluate how frequently the proposed dynamic transfer mechanism is activated.

$$TransferFrequency = \frac{\text{Total number of transfer events}}{\text{Number of simulation runs}} \tag{7}$$

- **Rescue-Time Benefit:** The rescue-time improvement gained by the higher-priority emergency vehicle after a transfer occurs. It is calculated as the difference between its rescue time under a baseline approach and its rescue time under the proposed SD-TLP multi-emergency vehicle approach using the same simulation seed, as shown in (8). A positive value means that the higher-priority emergency vehicle saved time under the proposed approach, while a negative value means that it experienced additional delay.

$$Benefit = ART^{HP}_{\text{baseline}} - ART^{HP}_{\text{proposed}} \tag{8}$$

- **Rescue-Time Cost:** The rescue-time cost imposed on the displaced lower-priority emergency vehicle after a transfer occurs. It is calculated as the difference between its rescue time under the proposed SD-TLP multi-emergency vehicle approach and its rescue time under a baseline approach using the same simulation seed, as shown in (9). A positive value means that the displaced lower-priority emergency vehicle experienced additional delay under the proposed approach, while a negative value means that it saved time.

$$Cost = ART^{LP}_{\text{proposed}} - ART^{LP}_{\text{baseline}} \tag{9}$$

In (8) and (9), HP refers to the higher-priority emergency vehicle used in the Rescue-Time Benefit metric, and LP refers to the displaced lower-priority emergency vehicle used in the Rescue-Time Cost metric.

- **Net Rescue-Time Delta:** The overall rescue-time effect of a transfer event on both emergency vehicles involved in the transfer. It is calculated by subtracting the Rescue-Time Cost from the Rescue-Time Benefit, as shown in (10). A positive value indicates that the transfer produced an overall rescue-time improvement. A negative value indicates that the delay imposed on the displaced emergency vehicle was greater than the benefit gained by the higher-priority emergency vehicle.

$$Net\ Delta = Benefit - Cost \tag{10}$$

### 7.5 Experimental Design

The simulation is configured using a set of parameters that define the experimental setup. These parameters are divided into varying parameters, which are adjusted to evaluate system performance, and fixed parameters, which remain constant throughout the experiments. Each configuration was evaluated under three traffic levels: low, medium, and high traffic, and each simulation run lasted 3000 s. A warm-up period of 300 s is used at the beginning of each simulation run to allow the traffic conditions to stabilize. Every configuration was executed using ten different random seeds. As a result, a total of 420 runs were conducted, and the final results were obtained by calculating the average of the ten runs.

### 7.5.1 Varying parameters

Varying parameters can either be intentionally modified to evaluate the performance of the system or change dynamically based on fixed parameters. The first varying parameter is the traffic arrival rate, which represents the flow of normal vehicles entering a road segment. Three traffic levels, including low, medium, and high, are considered in order to capture more realistic traffic conditions. In this study, low, medium, and high traffic arrival rates correspond to an average of one vehicle arriving every 10 s, 5 s, and 3.5 s, respectively, to simulate realistic traffic behavior. Based on the traffic arrival rate, the number of vehicles ahead of the emergency vehicle at an intersection, denoted as C, is obtained dynamically during the simulation run.

Next, the number of emergency vehicles is varied, where scenarios involve one, two, or three emergency vehicles. The estimated arrival time (EAT), as defined in Section 3, is used to resolve conflicts when emergency vehicles share the same priority level, where the vehicle with the smaller EAT is granted preemption first. The estimated time (ET) required to clear the queue at an intersection and the detection distance (DD), as defined in Section 3, are used to determine when traffic signal preemption should start. The varying parameters are summarized in Table 6.

Table 6: Varying Parameters

| **Parameter** | **Meaning** | **Value** |
|---|---|---|
| Traffic Arrival Rate | The rate at which vehicles enter the road network | Low (1 vehicle every 10 s), Medium (1 vehicle every 5 s), High (1 vehicle every 3.5 s) |
| Number of Emergency Vehicles | The number of emergency vehicles simultaneously present | 1–3 |
| EAT | The estimated travel time required for an emergency vehicle to reach the intersection | $\frac{D_{EV}}{S_{EV}}$ |
| $C$ | Number of vehicles on the road | Depends on arrival rate |
| ET | Estimated time required to clear the queue at an intersection | $\frac{C \cdot L + (C-1) \cdot MG}{S} + r$ |
| DD | Detection distance from the emergency vehicle to the traffic light | $ET \cdot S_{EV}$ |

### 7.5.2 Fixed parameters

Fixed parameters are constants that remain unchanged during the implementation and evaluation. These parameters are used with the simulation map described in Section 7.2.

After the emergency vehicle passes the current intersection, the Dthreshold, which is the distance between the current intersection and the previous intersection, is evaluated. If this distance exceeds 150 m, only the current traffic light is switched to red. Otherwise, when the distance is below 150 m, both traffic lights are switched to red. In this study, a value of 150 m is used for each intersection. The value is selected as an assumption based on half of the road segment length. The traffic load threshold value (TD threshold) used to activate traffic signal preemption is fixed at 0.7, as defined in [1]. This threshold is evaluated based on traffic conditions at the next two traffic signals along the emergency vehicle route, and preemption is triggered when the traffic load exceeds this value.

The average emergency vehicle speed ($S_{EV}$) is fixed at 60 km/h, which is the speed reported in [37], for emergency vehicles driving in urban areas. The average speed of normal vehicles (S) is set to 35 km/h, reflecting typical urban traffic conditions as stated in [39]. The same reference also states that the average vehicle length is 4.5 m. Following [1], the minimum gap between vehicles (MG), which represents a safe stopping distance in roads, is set to 2.5 m. The traffic light switching delay (r), representing the delay when switching between traffic signal phases, is set to 3 s, as done in [1]. Moreover, fixed traffic signal timing is applied at each four-way intersection. The full cycle length is approximately 120 s [39,40], which is divided equally among the four approaches, giving 30 s for each phase. Each phase includes 3 s of yellow and 2 s of all-red clearance, leaving 25 s for green. The 3 s yellow interval follows the MUTCD minimum yellow duration [41], while the 2 s all-red clearance is within the typical 1–3 s range for urban intersections [42]. From one approach's perspective, this results in 25 s green, 3 s yellow, 84 s red, and 8 s total all red clearance across the full cycle. These fixed timings are used when no emergency vehicle preemption is active. The fixed parameters and their corresponding values are listed in Table 7.

Table 7: Fixed Simulation Parameters

| **Parameter** | **Meaning** | **Value** |
|---|---|---|
| Road Segment Length | The length of each road segment | 300 m |
| $D_{threshold}$ | Threshold distance between two consecutive intersections | 150 m |
| TD threshold | Threshold used to determine whether traffic signal preemption should be activated based on traffic load conditions at the next two traffic signals | 0.7 |
| $S_{EV}$ | Average emergency vehicle speed | 60 km/h |
| MG | Minimum gap between vehicles | 2.5 m |
| $r$ | Traffic light switching delay | 3 s |
| $L$ | Average vehicle length | 4.5 m |
| $S$ | Average speed of regular vehicles | 35 km/h |
| Priority Rule | Priority order among emergency vehicles | Ambulance > Firetruck > Police |
| Traffic Signal Timing | Fixed signal phases used at each intersection | Green = 25 s, Yellow = 3 s, Red = 84 s (120 s cycle) |

### 7.5.3 Implementation and Validation

We implemented the SD-TLP algorithm of Bagheri et al. [1] in OMNeT++ version 6.3.0 following the specification given in Algorithm 1 in [1]. To validate our implementation, we ran the single-emergency-vehicle scenario (Scenario 1) under three traffic densities and verified that the Average Rescue Time (ART) matches the trends reported in [1]. However, a direct comparison of AWT and AIWT with the paper is not possible because the work does not specify the exact measurement window used for these metrics. The complete implementation of our work is available at:
https://github.com/Sarah0044/multiple-EV-SD-TLP-using-OMNeT-.git.

# 8. Results and Discussion

In this section, the results and discussion of the proposed multi-emergency vehicle SD-TLP system are presented and compared with the baselines: SD-TLP with FCFS, SD-TLP Closest that selects the closest emergency vehicle to the accident location, and No Preemption. Evaluation was conducted across four different scenarios. Scenario 1 considers a single emergency vehicle and is used to reproduce and compare results with [1], where the evaluation was performed only using the original SD-TLP algorithm and No Preemption. Scenarios 2 and 3 involve two emergency vehicles, while Scenario 4 involves three emergency vehicles. Each scenario was evaluated under three traffic conditions (low, medium, and high), and the reported results represent the average of ten simulation runs, along with the 95% confidence interval for the average ART, AWT, and AIWT metrics. All reported results assume ideal sensing and communication conditions, where the controller receives real-time information from RSUs without delay and packet loss. The results of each scenario are presented below alongside their corresponding discussion. The discussion mainly focuses on the performance of the emergency vehicles and the impact of the proposed approach on normal traffic. In addition, the Mann–Whitney U test was used in each scenario to evaluate whether the observed differences between the compared methods were statistically significant. This non-parametric test was selected because it does not assume normal distribution of the simulation results and is suitable for comparing independent samples with small sample sizes [43].

## 8.1 Scenario 1: Single Emergency Vehicle

Table 8: Scenario 1: Single Emergency Vehicle

| **Method** | **Traffic** | **ART (s)** | **AWT (s)** | **AIWT (s)** | **Ctrl. Over.** |
|---|---|---|---|---|---|
| SD-TLP | Low | $65.2 \pm 27.07$ | $30.25 \pm 15.30$ | $17.26 \pm 4.74$ | $4.0 \pm 0.0$ |
| SD-TLP | Medium | $76.4 \pm 30.87$ | $52.57 \pm 10.35$ | $29.14 \pm 4.77$ | $4.0 \pm 0.0$ |
| SD-TLP | High | $114.6 \pm 24.55$ | $55.32 \pm 11.13$ | $42.33 \pm 7.98$ | $5.8 \pm 0.87$ |
| No Preemption | Low | $113.4 \pm 24.70$ | $31.45 \pm 8.48$ | $16.74 \pm 2.00$ | $0.0 \pm 0.0$ |
| No Preemption | Medium | $135.1 \pm 33.28$ | $33.22 \pm 8.27$ | $18.99 \pm 3.56$ | $0.0 \pm 0.0$ |
| No Preemption | High | $528.0 \pm 480.72$ | $55.08 \pm 14.83$ | $39.63 \pm 7.78$ | $0.0 \pm 0.0$ |

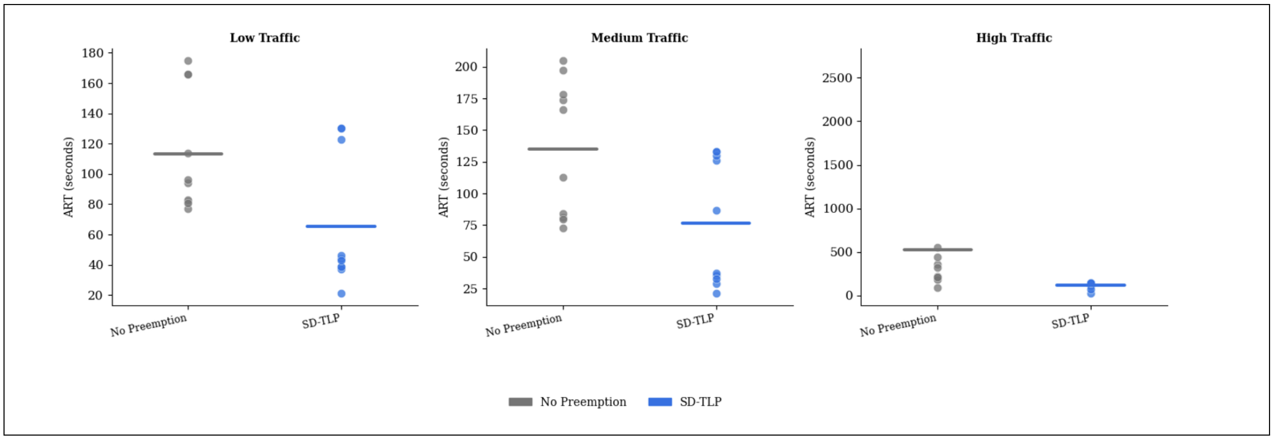



Figure 4: Scenario 1 per-seed scatter plot of ART, Dots= Individual Seeds, Horizontal Line= Mean

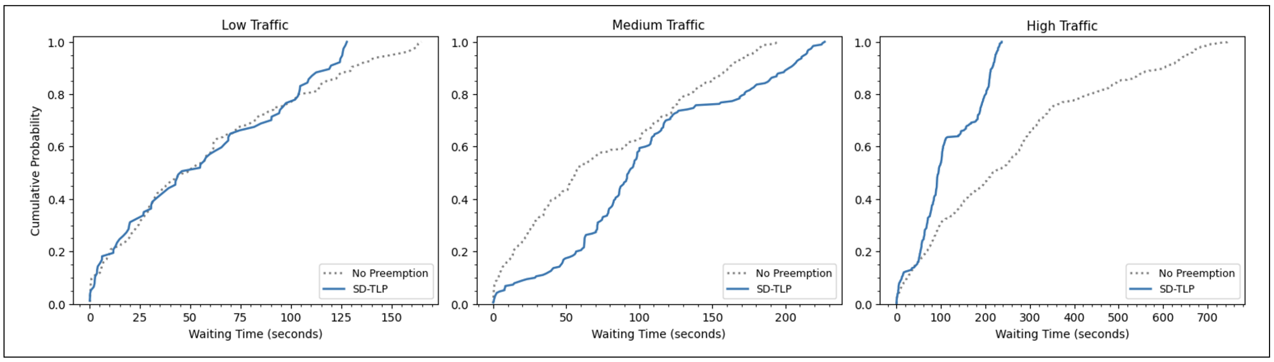



Figure 5: Scenario CDF of AWT for scenario 1

Table 9: Scenario 1: Mann–Whitney U Test Summary

| Metric | Comparison | Low | Medium | High |
|---|---|---|---|---|
| ART | SDTLP vs No Preemption | Sig. ($p = 0.031$) | Not Sig. ($p = 0.054$) | Sig. ($p = 0.0017$) |
| AWT | SDTLP vs No Preemption | Not Sig. ($p = 0.970$) | Sig. ($p = 0.009$) | Not Sig. ($p = 0.427$) |
| AIWT | SDTLP vs No Preemption | Not Sig. ($p = 0.850$) | Sig. ($p = 0.006$) | Not Sig. ($p = 0.345$) |

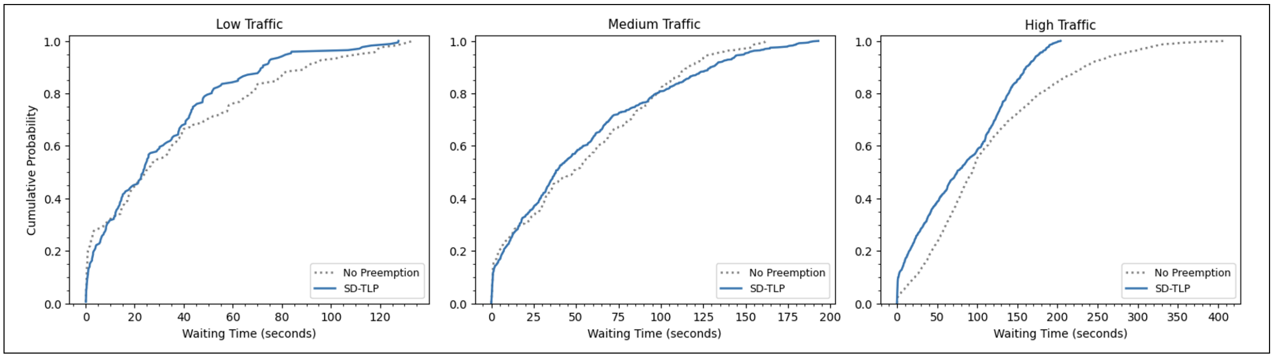



Figure 6: Scenario CDF of AIWT for scenario 1

## 8.2 Scenario 2: Two Emergency Vehicles (Different Priority)

Table 10: Scenario 2: Two Emergency Vehicles (Different Priority)

| Method | Traffic | EV1 ART (s) | EV2 ART (s) | Avg ART | AWT (s) | AIWT (s) | Ctrl. Over. |
|---|---|---|---|---|---|---|---|
| SD-TLP Multi | Low | 46.3 | 57.1 | $51.70 \pm 4.66$ | $20.88 \pm 10.16$ | $15.13 \pm 2.30$ | $5.6 \pm 0.43$ |
| SD-TLP Multi | Medium | 64.2 | 55.1 | $59.65 \pm 5.66$ | $33.96 \pm 13.83$ | $18.99 \pm 4.08$ | $5.9 \pm 0.54$ |
| SD-TLP Multi | High | 102.1 | 347.9 | $225.00 \pm 254.05$ | $52.87 \pm 13.51$ | $36.52 \pm 8.10$ | $10.1 \pm 3.26$ |
| SD-TLP FCFS | Low | 74.2 | 39.9 | $57.05 \pm 5.58$ | $37.21 \pm 11.29$ | $22.59 \pm 5.30$ | $5.8 \pm 0.39$ |
| SD-TLP FCFS | Medium | 129.4 | 61.1 | $95.25 \pm 22.52$ | $43.90 \pm 10.30$ | $23.64 \pm 4.73$ | $5.8 \pm 0.70$ |
| SD-TLP FCFS | High | 215.0 | 1060.5 | $637.75 \pm 386.65$ | $69.00 \pm 6.63$ | $50.76 \pm 6.70$ | $6.3 \pm 1.01$ |
| SD-TLP Closest | Low | 79.9 | 43.1 | $61.50 \pm 11.80$ | $35.84 \pm 9.66$ | $16.99 \pm 3.85$ | $6.2 \pm 0.70$ |
| SD-TLP Closest | Medium | 108.6 | 63.8 | $86.20 \pm 17.13$ | $53.09 \pm 8.82$ | $27.26 \pm 4.28$ | $6.0 \pm 0.83$ |
| SD-TLP Closest | High | 230.5 | 242.6 | $236.55 \pm 104.15$ | $69.67 \pm 9.93$ | $44.72 \pm 5.05$ | $7.9 \pm 1.86$ |
| No Preemption | Low | 115.9 | 79.8 | $97.85 \pm 10.18$ | $37.21 \pm 5.23$ | $20.64 \pm 1.74$ | $0.0 \pm 0.0$ |
| No Preemption | Medium | 130.4 | 139.9 | $135.15 \pm 38.89$ | $38.25 \pm 5.63$ | $22.38 \pm 2.77$ | $0.0 \pm 0.0$ |
| No Preemption | High | 452.9 | 1173.7 | $813.30 \pm 415.95$ | $66.20 \pm 10.50$ | $50.68 \pm 8.73$ | $0.0 \pm 0.0$ |

Table 11: Scenario 2: Summary of Mann–Whitney U Test Results

| Comparison | ART | AWT | AIWT |
|---|---|---|---|
| SDTLP-Multi vs No Preemption | Sig. under all traffic levels<br>Low: $p = 0.000182$<br>Med.: $p = 0.001315$<br>High: $p = 0.000183$ | Sig. only under low traffic<br>$p = 0.009108$ | Sig. under low and high traffic<br>Low: $p = 0.004586$<br>High: $p = 0.017257$ |
| SDTLP-Multi vs SDTLP-FCFS | Sig. under med. and high traffic<br>Med.: $p = 0.007285$<br>High: $p = 0.031209$ | Sig. only under low traffic<br>$p = 0.037635$ | Sig. only under high traffic<br>$p = 0.037635$ |
| SDTLP-Multi vs SDTLP-Closest | Sig. only under high traffic<br>$p = 0.045074$ | Sig. only under low traffic<br>$p = 0.034226$ | Sig. only under medium traffic<br>$p = 0.021134$ |

Scenario 2 evaluates the behavior of the system when two emergency vehicles with different priorities, EV1 (higher priority, ambulance) and EV2 (lower priority, police), approach the same intersection, creating a conflict. The results of the scenario evaluated under the proposed multi-emergency vehicle algorithm and the baseline approaches (SD-TLP with FCFS, SD-TLP Closest, and No Preemption) are shown in Table 10.

The results show that the proposed SD-TLP multi-emergency vehicle algorithm prioritizes the higher-priority emergency vehicle (EV1) over the lower-priority emergency vehicle (EV2), particularly under low and high traffic, confirming that the proposed approach serves the higher-priority emergency vehicle first. An exception occurs under medium traffic conditions, where the ambulances experience a slightly higher Average Rescue Time (ART) value than the police. This is mainly due to traffic conditions ahead of the ambulance, which may cause a delay even after winning preemption. Despite this, the ambulance still achieves lower ART compared to the baseline methods. This behavior occurs because the proposed algorithm allows the winner of the preemption process to change when a higher priority emergency vehicle enters the detection distance (DD). Therefore, the proposed multi-emergency vehicle SD-TLP algorithm can initially serve preemption to the lower-priority emergency vehicle until the higher-priority emergency vehicle is detected, after which preemption control is transferred to the higher-priority emergency vehicle.

In contrast, under the SD-TLP with FCFS baseline, the ambulance often experiences a higher ART than the police vehicle, as shown in Table 10. This occurs because FCFS assigns priority solely based on arrival order at the detection distance (DD) without considering the priority level of the emergency vehicle type. As a result, the lower-priority emergency vehicle may obtain preemption if it arrives first, forcing the higher-priority emergency vehicle to wait until the current preemption cycle is completed. Under SD-TLP Closest, proximity to the accident position rather than emergency vehicle type priority determines which vehicle is served. This means that an ambulance that is slightly farther away will be deprioritized in favor of a police vehicle that is slightly closer. This is shown in Table 10 , where under low and medium traffic, the police (EV2) has much lower ART values than the ambulance (EV1). Under the No Preemption mechanism, emergency vehicles follow the normal traffic signal operations, and no priority is given to them. As a result, both emergency vehicles experience significantly larger ART values, as shown in Table 10, since they must wait for regular signal cycles and traffic conditions.

When considering the average ART values, however, it can be observed that although the proposed multi-emergency vehicle SD-TLP algorithm achieves lower ART values across all traffic levels, these improvements are not necessarily statistically significant compared to all baseline methods under every traffic condition. The Mann—Whitney U test in Table 11 shows that the proposed solution's average ART values are statistically significant against No preemption under all traffic levels. This is expected since without preemption, emergency vehicles must wait for regular signal cycles. In addition, the proposed SD-TLP algorithm average ART is only significant against SD-TLP FCFS under medium and high traffic, and against SD-TLP Closest under high traffic. However, the statistical significance observed under high traffic when compared to SD-TLP Closest should be interpreted carefully, because the proposed approach has a wide confidence interval under high traffic conditions (±254.05 s), indicating variability between runs.

In addition, table 10 shows that most of the confidence intervals under low and medium traffic are close and overlapping between the methods, indicating that the ART difference is small.Under high traffic, however, the difference between the average ART values becomes much larger, especially compared to SD-TLP with FCFS and No Preemption. This is also reflected in the per-seed scatter plot shown in Figure 7, where several simulation runs under high traffic experience extremely large rescue times, particularly for SD-TLP with FCFS and No Preemption. Moreover, the wide confidence intervals under high traffic, particularly for SD-TLP with FCFS and No Preemption, also indicate instability due to severe traffic.

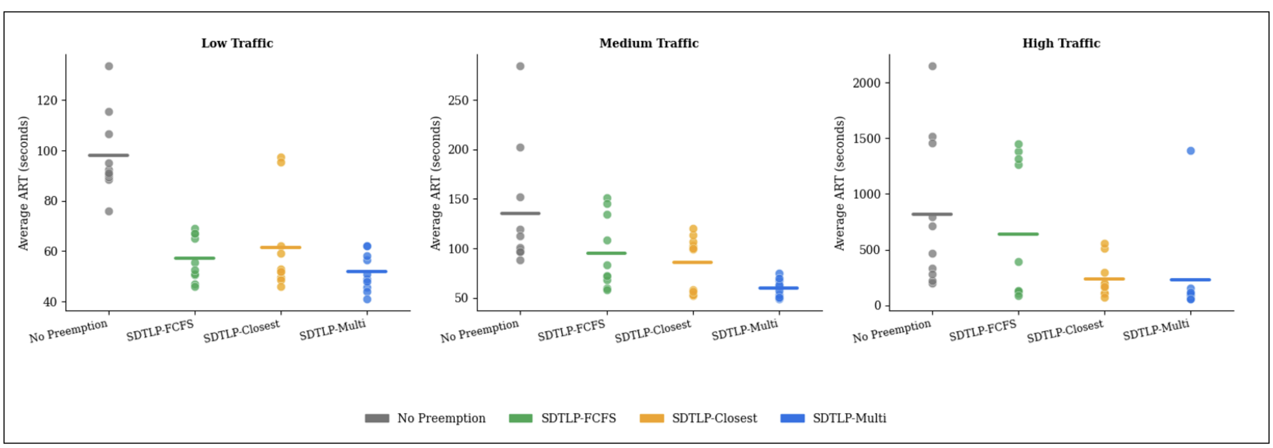



Figure 7: Scenario 2 per-seed scatter plot of average ART, Dots= Individual Seeds, Horizontal Line= Mean

Furthermore, in terms of Average Waiting Time (AWT) of normal vehicles, Table 10 shows that the proposed SD-TLP multi-emergency vehicle algorithm achieves the lowest AWT across the traffic conditions. This occurs because both emergency vehicles in the proposed approach are able to cross the intersection and reach their destination faster, reducing their impact on normal vehicles. However, the Mann—Whitney U test in Table 11 shows that the AWT of the proposed approach is only statistically significant under low traffic conditions.This behavior is also reflected in the cumulative distribution function (CDF) shown in Figure 8. In low traffic, the proposed multi-emergency vehicle SD-TLP approach curve rises faster than the baseline approaches, indicating that a large portion of normal vehicles experience lower waiting times when traveling on the same route as the emergency vehicles. Under medium and high traffic however, some overlaps between the curves exists, suggesting that most vehicles experience a similar waiting time, consistent with the non-significant AWT in Table 11. In addition, under high traffic, the No Preemption and SD-TLP Closest have a longer distribution tail, indicating that some normal vehicles experience very large waiting times, showing a pattern not visible from the AWT values in Table 10.

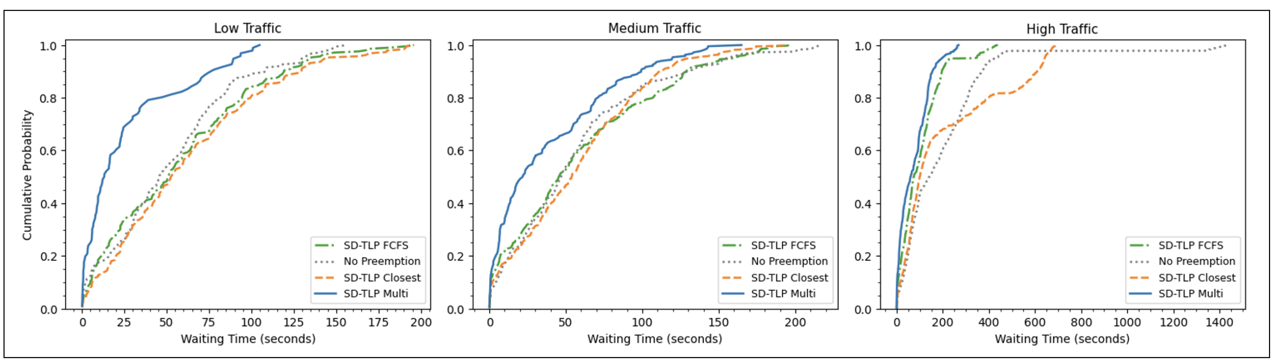



Figure 8: Scenario CDF of AWT for scenario 2

Similarly, the Average Imposed Waiting Time (AIWT) of normal vehicles is lower in the proposed approach compared to the baseline methods. This is mainly because, similar to AWT, both emergency vehicles in the proposed approach are able to cross the intersection and reach their destination faster, reducing their impact on normal vehicles. The statistical analysis in Table 11 shows that the results are significant under low and high traffic compared to No Preemption, under high traffic compared to SD-TLP FCFS, and under medium traffic compared to SD-TLP Closest. In addition, the CDF plot shown in Figure 9 provide further insights on the imposed waiting time of normal vehicles traveling in routes intersecting the emergency vehicles. Under low and medium traffic, the curve of the proposed SD-TLP approach rises faster than the baselines, indicating that a large portion of the normal vehicles in the intersecting routes experience lower waiting time. In addition, similar to AWT, some baseline methods, such as SD-TLP Closest in high traffic, have a longer distribution tail, indicating that some normal vehicles experience large waiting times, which is a pattern not directly visible from Table 10.


Figure 9: Scenario CDF of AIWT for scenario 2

### 8.3 Scenario 3: Two Emergency Vehicles (Same Priority)

Table 12: Scenario 3: Two Emergency Vehicles (Same Priority)

| Method | Traffic | EV1 ART (s) | EV2 ART (s) | Avg ART (s) | AWT (s) | AIWT (s) | Ctrl. Over. |
|---|---|---|---|---|---|---|---|
| SD-TLP Multi | Low | 62.1 | 46.8 | $54.45 \pm 8.02$ | $23.12 \pm 11.22$ | $16.04 \pm 3.09$ | $5.3 \pm 0.42$ |
| SD-TLP Multi | Medium | 76.4 | 50.8 | $63.60 \pm 11.72$ | $37.59 \pm 15.46$ | $21.49 \pm 4.74$ | $5.6 \pm 0.52$ |
| SD-TLP Multi | High | 105.5 | 345.9 | $225.70 \pm 253.22$ | $52.15 \pm 12.71$ | $36.37 \pm 7.67$ | $9.9 \pm 3.21$ |
| SD-TLP FCFS | Low | 92.6 | 38.4 | $65.50 \pm 10.29$ | $30.84 \pm 12.59$ | $19.94 \pm 4.51$ | $5.8 \pm 0.39$ |
| SD-TLP FCFS | Medium | 137.0 | 86.0 | $111.50 \pm 30.41$ | $42.49 \pm 11.25$ | $23.18 \pm 3.89$ | $5.4 \pm 0.84$ |
| SD-TLP FCFS | High | 239.2 | 1098.3 | $668.75 \pm 383.20$ | $67.27 \pm 9.46$ | $49.23 \pm 7.72$ | $6.8 \pm 2.06$ |
| SD-TLP Closest | Low | 85.1 | 53.7 | $69.40 \pm 12.07$ | $27.68 \pm 12.19$ | $15.61 \pm 1.50$ | $5.8 \pm 0.91$ |
| SD-TLP Closest | Medium | 101.1 | 86.6 | $93.85 \pm 21.22$ | $50.22 \pm 11.29$ | $27.60 \pm 4.79$ | $6.0 \pm 1.01$ |
| SD-TLP Closest | High | 163.7 | 286.2 | $224.95 \pm 103.43$ | $66.70 \pm 10.94$ | $40.69 \pm 4.43$ | $7.0 \pm 1.98$ |
| No Preemption | Low | 115.9 | 74.6 | $95.25 \pm 10.04$ | $35.99 \pm 5.43$ | $20.44 \pm 1.32$ | $0.0 \pm 0.0$ |
| No Preemption | Medium | 130.4 | 145.1 | $137.75 \pm 38.59$ | $36.74 \pm 5.75$ | $22.16 \pm 2.78$ | $0.0 \pm 0.0$ |
| No Preemption | High | 452.9 | 1167.6 | $810.25 \pm 415.98$ | $65.71 \pm 10.71$ | $50.85 \pm 8.76$ | $0.0 \pm 0.0$ |

Table 13: Scenario 3: Summary of Mann–Whitney U Test Results

| **Comparison** | **ART** | **AWT** | **AIWT** |
|---|---|---|---|
| SDTLP-Multi vs No Preemption | Sig. under all traffic levels<br>Low: $p = 0.000246$<br>Med.: $p = 0.000583$<br>High: $p = 0.001315$ | Sig. only under low traffic<br>$p = 0.037635$ | Sig. under low and high traffic<br>Low: $p = 0.007285$<br>High: $p = 0.025748$ |
| SDTLP-Multi vs SDTLP-FCFS | Sig. under med. and high traffic<br>Med.: $p = 0.005795$<br>High: $p = 0.01133$ | No sig. difference | No sig. difference |
| SDTLP-Multi vs SDTLP-Closest | Sig. under med. and high traffic<br>Med.: $p = 0.037492$<br>High: $p = 0.037635$ | No sig. difference | No sig. difference |

This scenario evaluates two emergency vehicles with the same vehicle-type priority approaching the same intersection from opposite directions, where the conflict is resolved mainly using the Estimated Arrival Time (EAT).

The results in Table 12 show that the proposed SD-TLP multi-emergency vehicle algorithm achieves the lowest average rescue time (ART) under low and medium traffic, and performs very close to SD-TLP Closest under high traffic. Under low traffic, the proposed approach achieves an average ART of 54.45 s, compared to 65.50 s for SD-TLP with FCFS, 69.40 s for SD-TLP Closest, and 95.25 s for No Preemption. A similar improvement is shown under medium traffic, where the proposed approach reduces the average ART to 63.60 s compared to 111.50 s, 93.85 s, and 137.75 s, respectively. Under high traffic, the proposed approach achieves an average ART of 225.70 s, which is very close to SD-TLP Closest with 224.95 s, and much lower than SD-TLP with FCFS and No Preemption. This improvement can be explained by the proposed algorithm checking again which emergency vehicle should receive preemption when another emergency vehicle becomes eligible. Therefore, instead of continuing to serve the first selected emergency vehicle, the controller can select the vehicle with the smaller EAT. This allows the emergency vehicle that is expected to reach the intersection sooner to receive preemption.

Looking at the individual ART values, EV2 achieves lower ART than EV1 under low and medium traffic, even though EV2 starts 6 seconds later. This is because the proposed approach can transfer preemption to EV2 when it enters the detection distance and has a smaller EAT. However, under high traffic, EV2 has a much larger ART than EV1. This suggests that heavy traffic on EV2's approach can still delay it even after preemption is given. Therefore, although the proposed approach handles the same-priority conflict using EAT, the final rescue time of each emergency vehicle still depends on the traffic condition on its approach.

Whereas, the SD-TLP with FCFS baseline selects the emergency vehicle that first enters the detection distance, even if the other emergency vehicle has a smaller EAT. As a result, FCFS may continue serving the first detected vehicle even when the other emergency vehicle could reach the intersection sooner. This leads to higher ART values, especially under medium and high traffic. Similarly, the SD-TLP Closest baseline selects the emergency vehicle based on the shortest remaining distance to the accident location. Although this can sometimes give balanced results, it does not consider the estimated arrival time to the intersection and does not transfer preemption when conditions change. Under No Preemption, both emergency vehicles follow the normal signal cycle, which leads to much larger rescue times across all traffic levels.

The Mann–Whitney U test in Table 13 shows that the ART difference between the proposed approach and No Preemption is statistically significant under all traffic conditions. In addition, the ART difference between the proposed approach and SD-TLP with FCFS is statistically significant under medium and high traffic, and the difference between the proposed approach and SD-TLP Closest is statistically significant under medium and high traffic. This means that the ART differences between these methods are unlikely to be due to random variation only. However, the direction of improvement should be interpreted using the average ART values in Table 12. Under high traffic, SD-TLP Closest has a slightly lower average ART than the proposed approach, with 224.95 s compared to 225.70 s. Therefore, although the difference is statistically significant, the average values show that the proposed approach is not always better than SD-TLP Closest under heavy congestion. This suggests that some transfer decisions under high traffic may reduce the rescue time of the selected emergency vehicle, but increase the rescue time of the emergency vehicle that loses preemption. Under low traffic, the difference compared with the SD-TLP baselines is not statistically significant. This can be explained by the lower congestion and shorter queues, where most preemption-based methods can already help the emergency vehicles pass efficiently.

Similar to Scenario 2, the confidence intervals show that the results are more consistent under low and medium traffic, while high traffic produces wider confidence intervals and larger differences between simulation runs. This is also shown in the per-seed scatter plot in Figure 10, where the high-traffic results are more spread out, especially for SD-TLP with FCFS and No Preemption. Similar to Scenario 2, the confidence intervals show that the results are more consistent under low and medium traffic, while high traffic produces wider confidence intervals and larger differences between simulation runs. This is also shown in the per-seed scatter plot in Figure 10, where the high-traffic results are more spread out, especially for SD-TLP with FCFS and No Preemption.

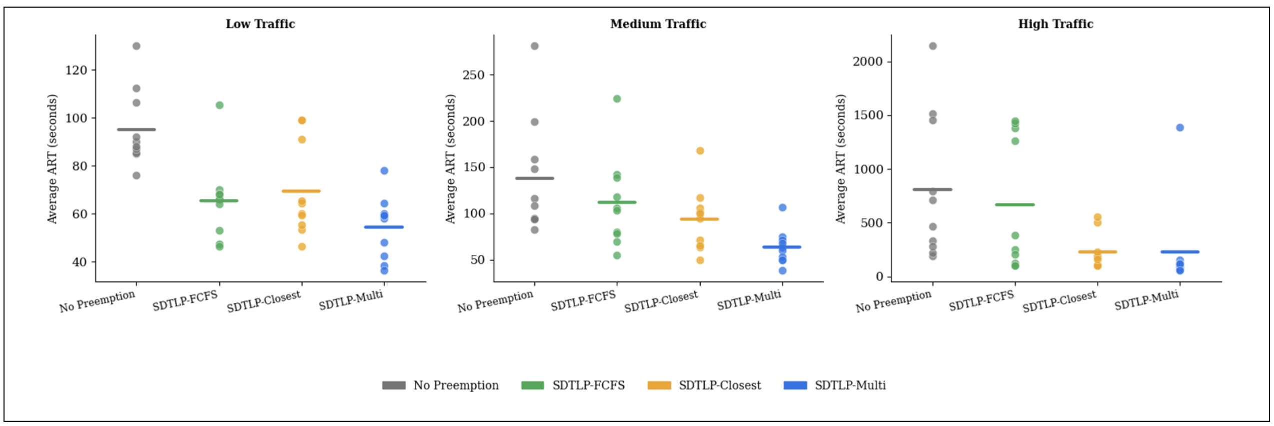



Figure 10: Scenario 3 per-seed scatter plot of average ART, Dots= Individual Seeds, Horizontal Line= Mean

As for the Average Waiting Time (AWT), Table 12 shows that the proposed SD-TLP multi-emergency vehicle algorithm achieves the lowest AWT under low and high traffic, and performs very close to No Preemption under medium traffic. Among the preemption-based methods, the proposed approach achieves the lowest AWT across all traffic conditions. This is because the proposed approach helps emergency vehicles clear the intersection more efficiently, which reduces the time that normal vehicles on the rescue route are affected. However, the Mann–Whitney U test in Table 13 shows that the AWT improvement is statistically significant only compared to No Preemption under low traffic. For the other SD-TLP baselines, the differences are not statistically significant. This means that although the proposed approach improves AWT numerically, the improvement for normal vehicles is smaller than the improvement for emergency vehicle rescue time.

This behavior is also supported by the CDF plot in Figure 11. Under low traffic, the proposed approach curve rises faster than the baseline approaches, which means that more normal vehicles on the rescue route have lower waiting times. Under medium and high traffic, the curves become closer to each other, which shows that the waiting time of normal vehicles becomes more similar between the methods as congestion increases.

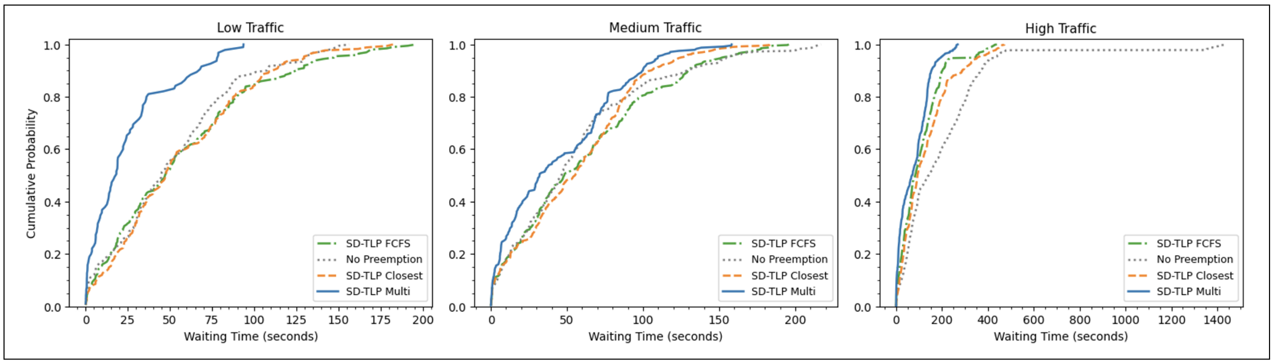



Figure 11: Scenario CDF of AWT for scenario 3

Similarly, the proposed approach achieves lower Average Imposed Waiting Time (AIWT) compared to most baseline methods across the traffic conditions. The statistical results in Table 13 show that the AIWT improvement is significant compared to No Preemption under low and high traffic, but not significant compared to SD-TLP with FCFS and SD-TLP Closest. This means that the proposed approach clearly improves performance compared to having no preemption. However, its effect on normal traffic delay compared with the other SD-TLP baselines is more limited.

The CDF plot in Figure 12 gives more details about the AIWT results. Under low and medium traffic, the proposed approach curve rises faster than the baseline methods, which means that many normal vehicles on intersecting routes experience lower imposed waiting time. Under high traffic, the curves become closer, showing that heavy congestion reduces the difference between the methods. However, some baseline methods still show longer tails, meaning that some normal vehicles experience very high waiting times.

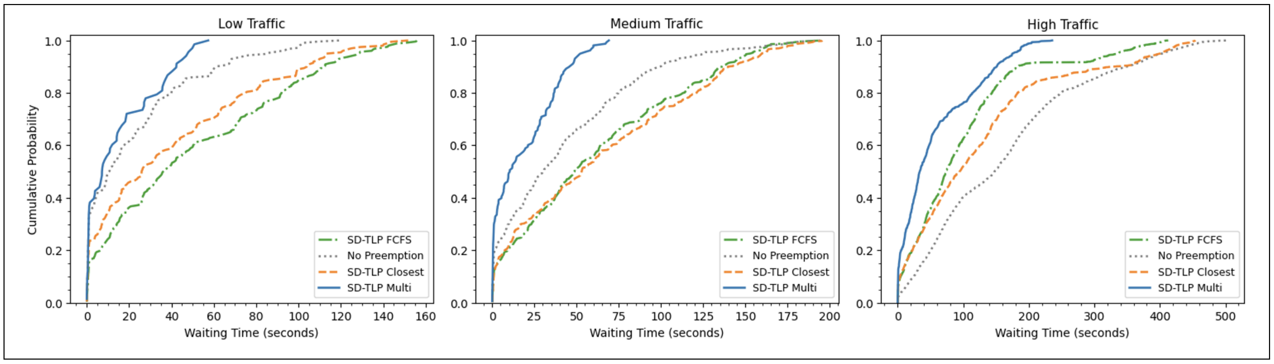



Figure 12: Scenario CDF of AIWT for scenario 3

Overall, Scenario 3 shows that when two emergency vehicles have the same priority, using EAT as the main conflict-resolution rule improves emergency vehicle rescue time, especially under medium and high traffic conditions. The proposed SD-TLP multi-emergency vehicle algorithm achieves the best average ART under low and medium traffic and performs very close to the best result under high traffic. It also achieves the lowest AWT among the preemption-based methods across all traffic levels. For AIWT, the proposed approach achieves the lowest value under medium and high traffic, while SD-TLP Closest is slightly lower under low traffic. However, the improvements in AWT and AIWT are not always statistically significant compared to the other SD-TLP baselines. This means that the main benefit of the proposed approach in this scenario is reducing emergency vehicle rescue time.

## 8.4 Scenario 4: Three Emergency Vehicles

Table 14: Scenario 4: Three Emergency Vehicles

| Method | Traffic | EV1 ART (s) | EV2 ART (s) | EV3 ART (s) | Avg ART (s) | AWT (s) | AIWT (s) | Ctrl. Over. |
|---|---|---|---|---|---|---|---|---|
| SD-TLP Multi | Low | 55.6 | 61.2 | 26.4 | $47.73 \pm 4.50$ | $29.67 \pm 9.99$ | $14.49 \pm 2.83$ | $6.6 \pm 0.60$ |
| SD-TLP Multi | Medium | 98.4 | 85.1 | 30.1 | $71.20 \pm 13.30$ | $38.43 \pm 10.20$ | $21.75 \pm 3.81$ | $7.8 \pm 1.46$ |
| SD-TLP Multi | High | 229.3 | 375.9 | 52.4 | $219.20 \pm 127.94$ | $61.84 \pm 11.67$ | $49.67 \pm 10.69$ | $13.9 \pm 2.90$ |
| SD-TLP FCFS | Low | 81.5 | 39.4 | 58.8 | $59.90 \pm 7.44$ | $34.88 \pm 11.43$ | $19.97 \pm 3.93$ | $7.2 \pm 0.64$ |
| SD-TLP FCFS | Medium | 119.8 | 55.4 | 99.3 | $91.50 \pm 21.49$ | $61.99 \pm 18.91$ | $26.22 \pm 4.32$ | $7.4 \pm 0.60$ |
| SD-TLP FCFS | High | 281.0 | 557.9 | 854.7 | $564.53 \pm 395.20$ | $68.96 \pm 10.56$ | $49.08 \pm 7.99$ | $8.1 \pm 2.01$ |
| SD-TLP Closest | Low | 94.6 | 47.8 | 36.3 | $59.57 \pm 12.46$ | $36.55 \pm 13.04$ | $19.16 \pm 5.43$ | $7.2 \pm 1.05$ |
| SD-TLP Closest | Medium | 90.0 | 69.8 | 47.9 | $69.23 \pm 7.97$ | $53.79 \pm 13.74$ | $24.43 \pm 5.77$ | $6.6 \pm 1.55$ |
| SD-TLP Closest | High | 212.1 | 260.9 | 366.0 | $279.67 \pm 240.66$ | $66.19 \pm 9.33$ | $43.22 \pm 7.52$ | $10.1 \pm 3.04$ |
| No Preemption | Low | 101.4 | 97.8 | 71.1 | $90.10 \pm 13.32$ | $43.93 \pm 7.79$ | $18.09 \pm 2.79$ | $0.0 \pm 0.0$ |
| No Preemption | Medium | 125.1 | 170.9 | 130.6 | $142.20 \pm 32.41$ | $46.36 \pm 4.83$ | $22.43 \pm 2.76$ | $0.0 \pm 0.0$ |
| No Preemption | High | 288.7 | 1744.3 | 2009.1 | $1347.37 \pm 395.80$ | $84.32 \pm 14.64$ | $67.49 \pm 11.54$ | $0.0 \pm 0.0$ |

Table 15: Scenario 4: Summary of Mann–Whitney U Test Results

| **Comparison** | **ART** | **AWT** | **AIWT** |
|---|---|---|---|
| SDTLP-Multi vs No Preemption | Sig. under all traffic levels<br>Low: $p = 0.000183$<br>Med.: $p = 0.001315$<br>High: $p = 0.00033$ | Sig. under low and high traffic<br>Low: $p = 0.017257$<br>High: $p = 0.045155$ | No sig. difference |
| SDTLP-Multi vs SDTLP-FCFS | Sig. only under low traffic<br>$p = 0.028306$ | No sig. difference | No sig. difference |
| SDTLP-Multi vs SDTLP-Closest | No sig. difference | No sig. difference | No sig. difference |

This scenario examines the performance of the system when three emergency vehicles approach the same intersection simultaneously, creating more complex preemption conflicts compared to the previous scenarios. In this scenario, EV1 and EV2 have the same priority level, while EV3 has a higher priority. Therefore, conflicts between EV1 and EV2 are mainly resolved using the EAT, whereas EV3 may receive preemption priority due to its higher priority level. The results of the scenario under the proposed approach and the baseline methods (SD-TLP with FCFS, SD-TLP Closest, and No Preemption) are presented in Table 14.

The results indicate that the proposed SD-TLP multi-emergency vehicle algorithm achieves the lowest ART under low traffic conditions and significantly outperforms the No Preemption approach under all traffic levels. Under low traffic, the proposed approach achieves an average ART of 47.73 s, compared to 59.90 s for SD-TLP FCFS, 59.57 s for SD-TLP Closest, and 90.10 s for No Preemption. Under medium traffic conditions, the proposed approach achieves an average ART of 71.20 s, while SD-TLP FCFS, SD-TLP Closest, and No Preemption achieve 91.50 s, 69.23 s, and 142.2 s, respectively. Although SD-TLP Closest achieves a slightly lower ART under medium traffic, the proposed approach still outperforms SD-TLP FCFS and No Preemption.

Under high traffic conditions, the average ART values increase significantly for all approaches due to severe congestion. The proposed approach achieves an average ART of 219.2 s, compared to 564.53 s for SD-TLP FCFS, 279.67 s for SD-TLP Closest, and 1347.37 s for No Preemption. These results demonstrate that the proposed dynamic preemption transfer mechanism becomes increasingly beneficial under highly congested traffic conditions.

Unlike the baseline approaches, the proposed mechanism continuously re-evaluates the traffic situation and can dynamically transfer preemption when a higher-priority emergency vehicle approaches the intersection.

Examining the individual rescue times helps explain the behavior of the proposed approach. Since EV1 and EV2 have the same priority level, conflicts between them are mainly resolved using EAT. Therefore, the emergency vehicle expected to reach the intersection sooner may receive preemption first. As a result, EV1 sometimes achieves lower ART values, while EV2 achieves lower ART values in other cases. In contrast, EV3 has a higher priority level, allowing it to receive preemption priority whenever conflicts occur with EV1 or EV2. This demonstrates that the proposed algorithm does not rely only on arrival order, but instead dynamically considers both emergency vehicle priority and EAT when allocating preemption. For example, under low traffic, EV3 achieves an ART of 26.4 s compared to 55.6 s for EV1 and 61.2 s for EV2. A similar behavior can be observed under medium traffic, where EV3 achieves 30.1 s while EV1 and EV2 achieve 98.4 s and 85.1 s, respectively. Under high traffic, EV3 still maintains the lowest rescue time among the three emergency vehicles with 52.4 s, compared to 229.3 s for EV1 and 375.9 s for EV2.

In contrast, SD-TLP FCFS allocates preemption according to the order in which emergency vehicles enter the detection distance, without considering updated traffic conditions, EAT, or emergency vehicle priorities. This means that a lower priority emergency vehicle may receive preemption before another vehicle that could reach the intersection first. Similarly, SD-TLP Closest mainly selects the emergency vehicle with the shortest remaining distance to the accident location. Although this strategy may improve rescue performance in some situations, it does not dynamically adapt to changing traffic conditions or estimated arrival times. Under the No Preemption mechanism, all emergency vehicles follow the normal traffic signal operation, resulting in substantially larger rescue times under all traffic conditions.

The Mann–Whitney U test results presented in Table 15 further support these observations. The ART improvement achieved by the proposed approach is statistically significant compared to No Preemption under all traffic conditions, with p-values of 0.000183, 0.001315, and 0.00033 under low, medium, and high traffic, respectively. In addition, statistically significant improvements are observed compared to SD-TLP FCFS under low traffic conditions with a p-value of 0.028306. However, no statistically significant difference is observed between the proposed approach and SD-TLP Closest, indicating that both methods can provide competitive rescue performance in some cases.

The per-seed scatter plot shown in Figure 13 further illustrates the ART behavior across the different traffic conditions. Under low traffic, the results are relatively close together for all methods, indicating more stable rescue performance between simulation runs. Under medium traffic, larger variations begin to appear, particularly for No Preemption and SD-TLP FCFS. Under high traffic, the scatter becomes much wider, especially for No Preemption, SD-TLP FCFS, and SD-TLP Closest, indicating higher variability and less stable rescue performance under severe congestion. In contrast, the proposed SD-TLP Multi approach generally maintains lower ART values and fewer extreme outliers across the traffic conditions.

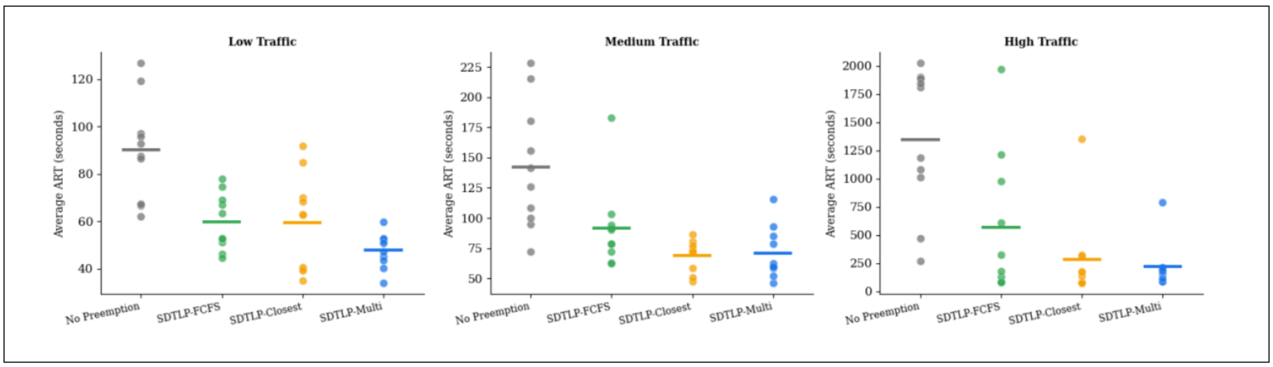



Figure 13: Scenario 4 per-seed scatter plot of average ART, Dots= Individual Seeds, Horizontal Line= Mean

Regarding the AWT of normal vehicles, the proposed approach generally achieves lower AWT values than most baseline approaches across the different traffic conditions. The proposed approach achieves AWT values of 29.67s, 38.43s, and 61.84s under low, medium, and high traffic conditions, respectively, as shown in Table 14. This occurs because the proposed approach helps emergency vehicles clear intersections more efficiently, reducing traffic disruption along the rescue route. The statistical analysis in Table 15 shows that the AWT improvement achieved by the proposed approach is statistically significant compared to No Preemption under low and high traffic conditions, with p-values of 0.017257 and 0.045155, respectively. However, no statistically significant difference is observed compared to SD-TLP FCFS or SD-TLP Closest. This suggests that although the proposed approach reduces the waiting time of normal vehicles, its main advantage is improving emergency vehicle rescue time.

The CDF plots shown in Figure 14 further support these results. Under low traffic conditions, the proposed approach has a steeper curve than the baseline methods, indicating that more vehicles experience shorter waiting times. Under medium and high traffic conditions, the curves become closer together as congestion increases and reduces the differences between the methods. However, some baseline approaches still show longer tails, meaning that some vehicles experience very large waiting times, which occurs less often with the proposed approach.

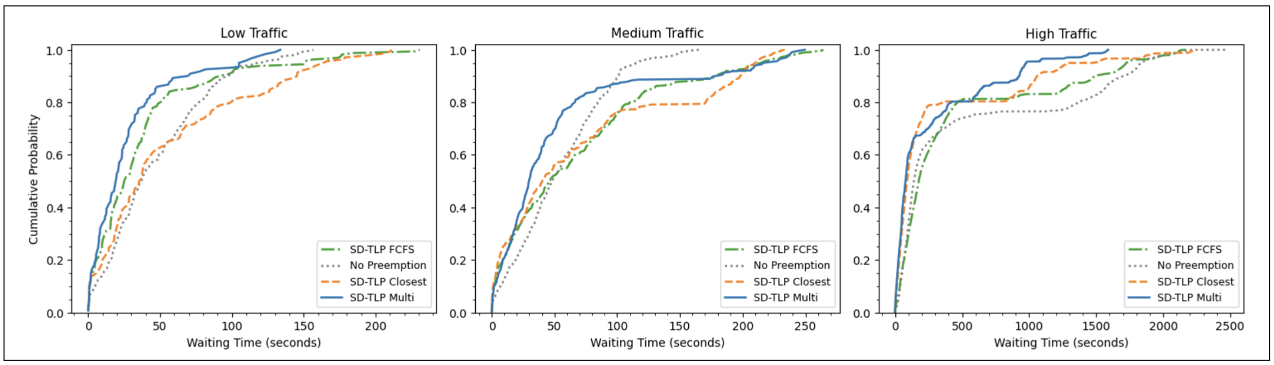



Figure 14: Scenario CDF of AWT for scenario 4

Similarly, the AIWT results in Table 14 show that the proposed approach achieves competitive performance compared to the baseline methods. It achieves AIWT values of 14.49 s, 21.75 s, and 49.67 s under low, medium, and high traffic conditions, respectively. The proposed approach reduces the waiting time of vehicles traveling on routes intersecting the emergency vehicle path by improving the efficiency of emergency vehicle movement through intersections. However, the Mann–Whitney U test results in Table 15 show that the AIWT differences are not statistically significant compared to the baseline approaches. This suggests that although the proposed approach can reduce imposed waiting time in some cases, the improvement is small.

The AIWT CDF plots shown in Figure 15 provide more details about the waiting times of affected vehicles. Under low traffic conditions, the proposed approach shows a steeper curve, meaning that more vehicles experience shorter imposed waiting times. Under medium and high traffic conditions, the curves become more similar because heavy congestion reduces the differences between the methods. However, some baseline approaches still show longer tails, meaning that some vehicles experience very large waiting times, which happens less often with the proposed approach.

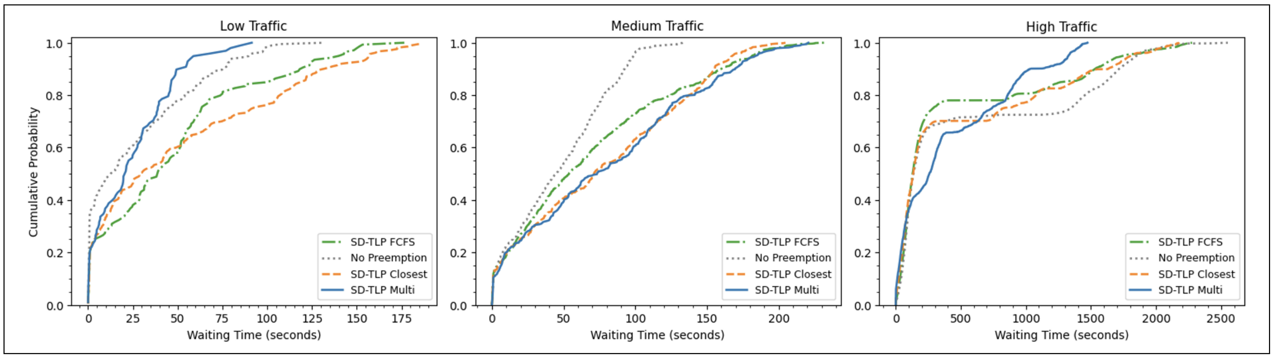



Figure 15: Scenario CDF of AIWT for scenario 4

Overall, Scenario 4 demonstrates that the proposed SD-TLP multi-emergency vehicle algorithm can effectively manage more complex conflicts involving three simultaneous emergency vehicles by considering both emergency vehicle priority and EAT. The proposed approach consistently achieves lower rescue times than No Preemption and outperforms SD-TLP FCFS under all traffic conditions while remaining competitive with SD-TLP Closest. In addition, the proposed approach maintains competitive AWT and AIWT values, showing that the approach can improve emergency response performance while limiting the impact on normal traffic vehicles.

### 8.5 Characterization of Dynamic Preemption Transfer

This section evaluates the dynamic transfer mechanism in the proposed SD-TLP multi-emergency vehicle approach. In this mechanism, when an emergency vehicle with a higher priority (based on the rules in Section 6.3 ) enters its detection distance (DD), preemption is transferred to that emergency vehicle. Figure 16 shows an example of a run in Scenario 3, where preemption was changed from EV0 to EV1 due to EV1 having a shorter EAT to the intersection.

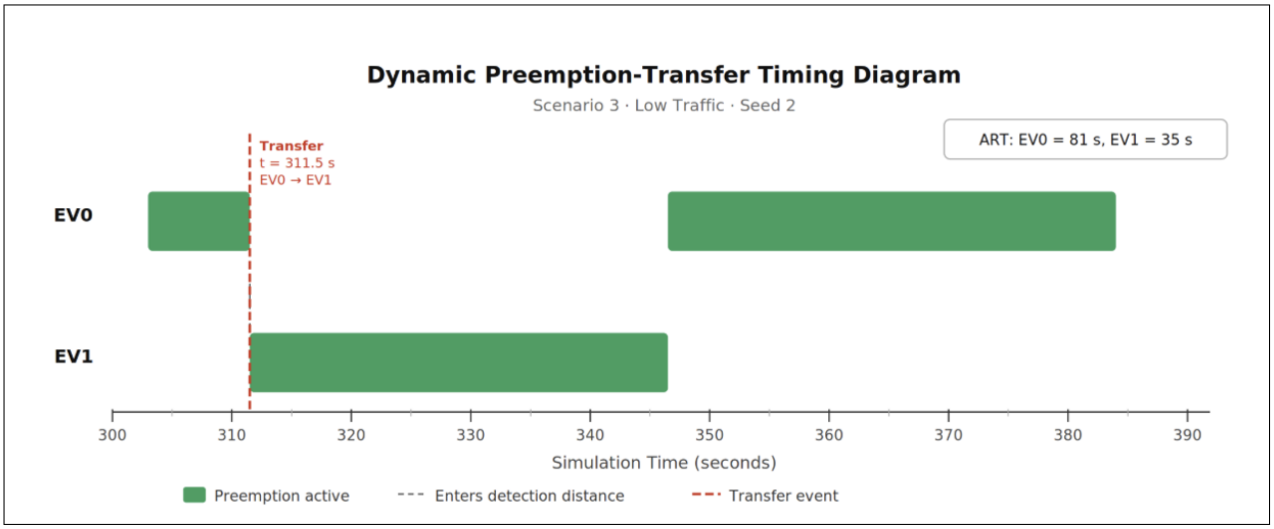



Figure 16: Time-series of preemption events for seed 2 in scenario 3

In addition, tables 16 and 17 present the dynamic transfer results when comparing the proposed approach against the baselines SD-TLP Closest and SD-TLP with FCFS, respectively.

Table 16: Dynamic Preemption-Transfer Characterization — SD-TLP Multi vs. SD-TLP Closest

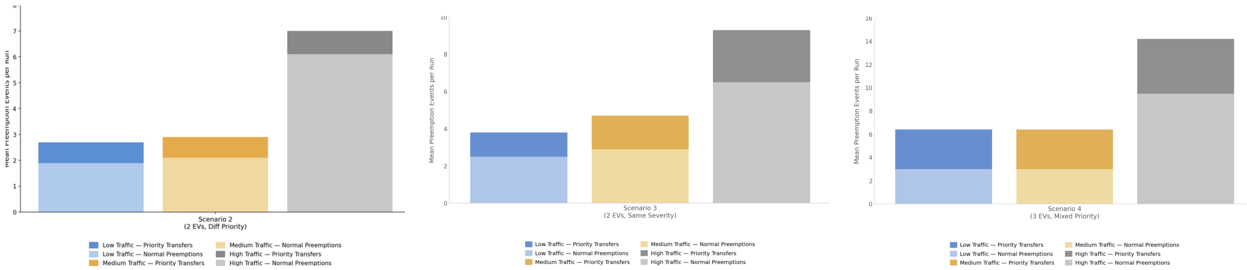
| Scenario | Traffic | Transfers/Run | Rescue-Time Benefit to Winner (s) | Rescue-Time Cost to Displaced (s) | Net Rescue-Time Delta (s) |
|---|---|---|---|---|---|
| Sc. 2 (2 EVs, Diff. Priority) | Low | 0.9 | 17.9 s | 10.2 s | 7.7 s |
| Sc. 2 (2 EVs, Diff. Priority) | Medium | 0.9 | 64.8 s | 1.2 s | 63.6 s |
| Sc. 2 (2 EVs, Diff. Priority) | High | 1 | 170 s | 456.2 s | -286.2s |
| Sc. 3 (2 EVs, Same Priority) | Low | 1.3 | 12.5 s | -14.7s | 27.2 s |
| Sc. 3 (2 EVs, Same Priority) | Medium | 1.8 | 36.1 s | -23.9 s | 60.0 s |
| Sc. 3 (2 EVs, Same Priority) | High | 2.8 | -140 s | 100.6 s | -241.1 s |
| Sc. 4 (3 EVs, Mixed Priority) | Low | 3.4 | 14.3s | -12.5s | 26.9s |
| Sc. 4 (3 EVs, Mixed Priority) | Medium | 3.4 | 3.6s | 13.3s | -9.6s |
| Sc. 4 (3 EVs, Mixed Priority) | High | 4.9 | 111.2s | 31.3s | 79.9s |

Table 17: Dynamic Preemption-Transfer Characterization — SD-TLP Multi vs. SD-TLP FCFS

| Scenario | Traffic | Transfers/Run | Rescue-Time Benefit to Winner (s) | Rescue-Time Cost to Displaced (s) | Net Rescue-Time Delta (s) |
|---|---|---|---|---|---|
| Sc. 2 (2 EVs, Diff. Priority) | Low | 0.9 | 21.6 s | 9.2 s | 12.3 s |
| Sc. 2 (2 EVs, Diff. Priority) | Medium | 0.9 | 109.6 s | -0.1 s | 109.7 s |
| Sc. 2 (2 EVs, Diff. Priority) | High | 1 | 133 s | -879.5 | 1012.5 s |
| Sc. 3 (2 EVs, Same Priority) | Low | 1.3 | 13.5 s | -14.2 s | 27.6 s |
| Sc. 3 (2 EVs, Same Priority) | Medium | 1.8 | 51.7 s | -36.4 s | 88.1 s |
| Sc. 3 (2 EVs, Same Priority) | High | 2.8 | 564.1 s | -257.3 s | 821.4 s |
| Sc. 4 (3 EVs, Mixed Priority) | Low | 3.4 | 9.9 s | -3.4 s | 13.3 s |
| Sc. 4 (3 EVs, Mixed Priority) | Medium | 3.4 | 31.7 s | -7.2 s | 38.9 s |
| Sc. 4 (3 EVs, Mixed Priority) | High | 4.9 | 444.0 s | -72.2 s | 516.3 s |

The results in Tables 16 and 17 and Figure 17 show that the mean frequency of dynamic preemption transfer increases as both traffic density and the number of concurrent emergency vehicles increase. In Figure 17, normal preemption refers to the case where an emergency vehicle receives preemption control, while priority-transfer event refers to the case where the currently served emergency vehicle is replaced by another emergency vehicle. Scenario 2 produces the lowest transfer frequency, with approximately one transfer event per run, while scenario 4 produces the highest transfer frequency, reaching 4.9 transfers per run under high traffic conditions. In scenario 2, the transfer frequency remains low because EV1(ambulance) always has higher priority than EV2 (police) due to the emergency vehicle type priority. Therefore, transfer typically occurs only once when the higher-priority emergency vehicle enters the detection distance (DD) and takes control of the preemption process. In contrast, scenario 3 involves two emergency vehicles of the same type, causing the controller to repeatedly re-evaluate the winner based on estimated arrival time (EAT) to the intersection and distance to the accident position when they have equal EAT values, resulting in more transfer events. In addition, scenarios involving a larger number of emergency vehicles, such as scenario 4, create more conflict for preemption control, increasing the likelihood of transfer events. Furthermore, Figure 17 shows that normal preemptions usually occur more frequently than priority transfers, because not every preemption process requires reassignment to another emergency vehicle.

When considering the per-event rescue-time cost and benefit shown in Tables 16 and 17, the results show that the proposed transfer mechanism consistently produces positive net rescue-time deltas when compared to SD-TLP with FCFS across all scenarios and traffic conditions. This occurs because SD-TLP with FCFS serves the initially selected emergency vehicle based only on arrival order, where the emergency vehicle that enters the detection

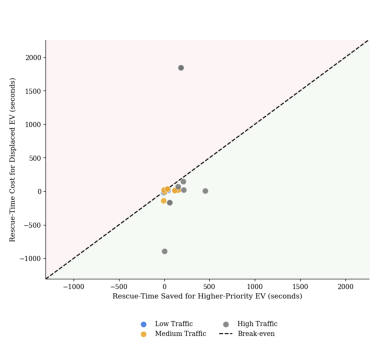



(a) Scenario 2 (b) Scenario 3 (c) Scenario 4

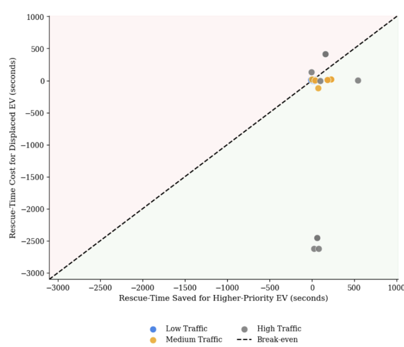
Figure 17: Stacked bar charts of the mean number of normal preemptions and priority-transfer events per simulation run under different traffic conditions.

distance (DD) is selected first, even when a higher priority emergency vehicle later requests preemption. As a result, the transfer mechanism is able to reduce rescue-times of emergency vehicles, particularly under medium and high traffic conditions where congestion increases the impact of selecting a less suitable emergency vehicle.

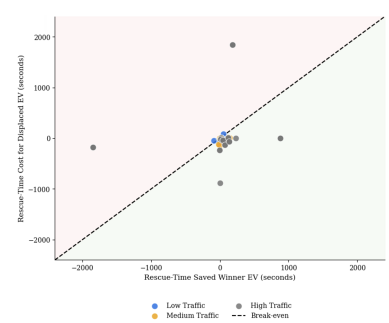
In contrast, the comparison against SD-TLP Closest produces more mixed results. This is because SD-TLP Closest already selects the emergency vehicle that is geographically closer to the accident position, meaning that the initially selected emergency vehicle is often a reasonable choice for preemption. Therefore, the overall net rescue-time delta achieved by the transfer is small in some cases and may be negative, meaning that transfer is not beneficial. Under low traffic in scenarios 2,3, and 4, and under medium traffic in scenarios 2 and 3, the transfer mechanism achieves positive net rescue-time deltas. This means that the benefit to the newly selected emergency vehicle is greater than the cost imposed on the displaced emergency vehicle. However, under scenario 2 and scenario 3 in high traffic conditions, the net rescue-time delta becomes negative. This means that the rescue-time cost imposed on the displaced emergency vehicle becomes greater than the rescue-time benefit achieved by the newly selected emergency vehicle. This behavior is further illustrated in the benefit-versus-cost scatter plots as shown in Figure 18, where some transfer events under high traffic have a larger displaced (lower priority ) emergency vehicle cost compared to the rescue-time benefit achieved by the newly selected (higher priority) emergency vehicle. This occurs because under high congestion, the rescue-time cost experienced by the displaced (lower priority) emergency vehicle becomes larger than the rescue-time benefit achieved by the newly selected (higher priority) emergency vehicle. In addition, scenario 3 under high traffic shows a case where both the higher priority (newly selected) emergency vehicle and the displaced emergency vehicle experience rescue-time delays compared to SD-TLP Closest. This may indicate that under some traffic conditions and when multiple emergency vehicles compete for preemption, the transfer can sometimes introduce additional delays that affect both emergency vehicles when compared to SD-TLP Closest. In contrast, several low and medium traffic conditions, particularly under scenarios 3 and 4, show cases where both the newly selected emergency vehicle and the displaced emergency vehicle achieve rescue-time improvements compared to baseline methods. This suggests that under low and medium traffic, dynamic transfer may sometimes improve overall emergency vehicle rescue-times rather than simply causing delays for one emergency vehicle.



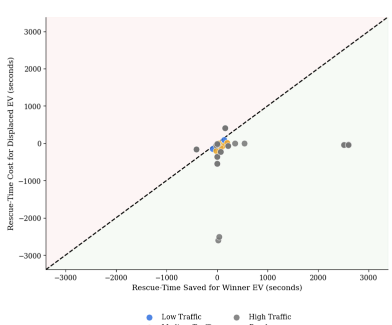

(a) Scenario 2 vs SD-TLP Closest



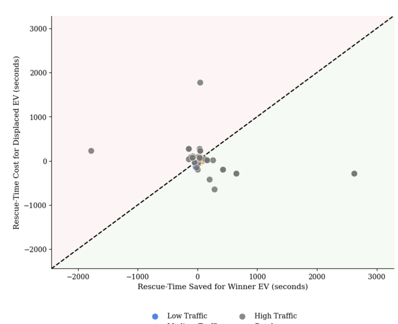

(b) Scenario 2 vs SD-TLP FCFS



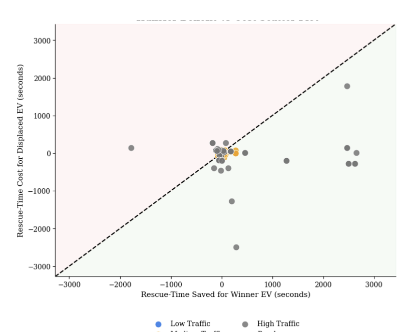

(c) Scenario 3 vs SD-TLP Closest




(d) Scenario 3 vs SD-TLP FCFS




(e) Scenario 4 vs SD-TLP Closest




(f) Scenario 4 vs SD-TLP FCFS

Figure 18: Per transfer event benefit-versus-cost scatter plots showing rescue-time benefit achieved by the newly selected emergency vehicle versus rescue-time cost experienced by the displaced emergency vehicle under different scenarios and baseline methods. Points below the break-even line represent transfer events where the rescue-time benefit achieved by the newly selected emergency vehicle is larger than the rescue-time cost experienced by the displaced emergency vehicle, while points above the line represent net negative transfer events.

Overall, the results suggest that the effectiveness of the dynamic preemption transfer mechanism depends strongly on the traffic conditions and the scenarios. Under most low and medium traffic conditions, the transfer mechanism produces positive net rescue-time deltas because the rescue time benefit achieved by the newly selected (higher priority) emergency vehicle is larger than the rescue-time cost experienced by the displaced emergency vehicle. However, under high traffic conditions, particularly in scenario 2 and scenario 3 when compared against SD-TLP Closest, the net rescue-time delta becomes negative. This suggests that when there is high congestion, transfers may introduce larger rescue-time costs for the displaced emergency vehicle than the rescue-time benefit achieved by the newly selected emergency vehicle. Furthermore, in scenario 2, the large negative net rescue-time delta observed suggests that although the proposed SD-TLP multi-emergency vehicle achieved lower average ART values shown in Table 10, some individual transfer events become inefficient under high traffic, affecting the overall net benefit of the transfer, as shown by the few outlier points in the scatter plots in Figure 18. Therefore, these results indicate that the dynamic preemption mechanism is most effective under low and medium traffic conditions, and becomes less reliable under high traffic conditions when compared to SD-TLP Closest.

### 8.6 Ethical Considerations

In addition to evaluating our system there are also ethical considerations that must be taken into account. One of the first priority rules when it comes to selecting the higher-priority emergency vehicle in our algorithm is based on the emergency vehicle type, which raises important ethical considerations. Although several other criteria can be used to determine priority, following the future work directions suggested in [1], we adopt the vehicle type priority scheme proposed in [2], where ambulances are assigned the highest priority, followed by fire trucks, and then police vehicles. The authors justified this ordering by stating it is based on the global driving rules. Additionally, from an ethical standpoint, ambulances are assigned the highest priority because they are usually the most involved in time-critical, life-threatening medical emergencies. In many health incidents, patient survival is highly dependent on the response time of the ambulance. For example, a delay of just one minute in cardiac arrest cases may decrease the chance of survival by 7-10%, as reported by the American Heart Association [44]. Similarly, in stroke emergencies, fast medical intervention is extremely important in improving health outcomes [45]. Therefore, fast ambulance arrival is essential, and in many cases, delays can lead to irreversible harm or even death. As for fire trucks, they are also critically important, and some may consider them as the first priority. However, according to the national fire incident data in the United States [46], only 3.9% of fire department dispatches in 2020 are fire related, while a significantly larger portion involved non-emergency or low-urgency incidents, such as animal rescue. Although fires can spread quickly, they usually take time to develop and cause harm. Therefore, fire trucks are placed as the second priority. Police vehicles are assigned the lowest overall priority among the three emergency vehicle types. The main reason for this is that, when compared to ambulances and fire trucks, most police responses do not involve life-threatening and time-critical scenarios, and only a minority of cases require immediate intervention [47]. It is also important to note that the proposed order of emergency vehicle type priority followed in this project may change depending on the area. For example, some may prioritize fire trucks over ambulances, and we may adapt this to our project accordingly.

# 9. Threats to Validity and Limitations

There are several threats to validity and limitations encountered in this work that should be acknowledged, including reproducibility threats, scenario realism, and statistical threats.

- **Reproducibility Threats:** There are several reproducibility threats in this work. First, the implementation relies solely on OMNeT++ version 6.3.0, in contrast to Bagheri et al, who used an unspecified version of OMNeT++, SUMO, and Veins. In addition, multiple parameter settings were not provided in [1], such as the maximum number of normal vehicles per approach and the traffic signal cycle operation. The source code was also not available, limiting reproducibility and requiring several assumptions to be made during implementation. Furthermore, it is important to note that the results depend on the selected scenarios and parameter settings, and different configurations may yield different results. In this work, we tried to consider several scenarios where conflicts may occur, given the timeframe.

- **Scenario Realism:** Since the implementation was performed using only OMNeT++, traffic and vehicle movements were modeled within the simulator as a simplified abstraction rather than being generated through an external traffic simulator such as SUMO integrated with Veins. This is considered one of the main threats to validity in this work, since removing SUMO and Veins changes the realism of the traffic model and the communication environment compared to the framework used by Bagheri et al. [1]. Moreover, the current simulation environment may simplify real-world conditions, especially since entities such as pedestrians and unpredictable drivers were not explicitly modeled.

  In addition, one of the main threats to validity is that the proposed mechanism assumes ideal sensing and communication conditions, where the centralized controller receives real-time information from roadside units (RSUs) without communication delay or packet loss. In real-world deployments, communication latency may affect the timing of traffic light preemption decisions and dynamic preemption transfer operations.

- **Statistical Threats:** The obtained results are based on a limited number of simulation runs (10 runs only for each scenario), which may not be enough to capture the movement of normal vehicles under low, medium, and high traffic.

- **Generalization Limitations:** The proposed mechanism was evaluated using four scenarios involving up to three emergency vehicles within a symmetric four intersection topology. Therefore, the results obtained may not generalize to larger road networks, asymmetric intersection topologies, or scenarios involving more than three emergency vehicles.

# 10. Conclusion

# 11. References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## Appendix A — Reproducibility Checklist

Below, the necessary information to reproduce the project is presented.

| | |
|---|---|
| Repository link | `https://github.com/Sarah0044/multiple-EV-SD-TLP-using-OMNeT-` |
| How to install dependencies | First, OMNeT++ 6.3.0 must be installed from their website. Next, import the project into the OMNeT IDE and build the project (right-click on the project then select "Build"). |
| How to run baseline experiments | To run baseline experiments, the **omnetpp.ini** file must be selected. Then, select the desired scenario you want to run from the Run configuration button. All the methods (the baselines for that scenario and the main algorithm) will be generated automatically. |
| How to run your extension experiments | To run the extended experiments, the **omnetpp.ini** file must be selected. Then, select the desired scenario you want to run from the Run configuration button. All the methods (the baselines for that scenario and the main algorithm) will be generated automatically. |
| Where results are stored | The simulation outputs are stored in the **results** folder |
| Random Seeds/Number of runs | Ten runs are conducted using seeds {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}. |

## Appendix B — AI Assistance Statement

We acknowledge the use of **Claude** and **ChatGPT** in assisting with the code and simulator use.

Additionally, we acknowledge using **ChatGPT** and **Grammarly** for writing assistance, including spelling and grammar correction.

## Appendix C — Additional Results Figures and Tables

This appendix presents additional tables and figures related to the experimental results and evaluation performed in this project.

### Per-Seed Scatter Plots of AWT and AIWT

The following sections show the per-seed scatter plots of the average waiting time (AWT) and average imposed waiting time (AIWT) per scenario.

### Scenario 1: Single Emergency Vehicle

Figures 19 and 20 show the per seed scatter plots of the average waiting time (AWT) and average imposed waiting time (AIWT) of normal vehicles in scenario 1, respectively.

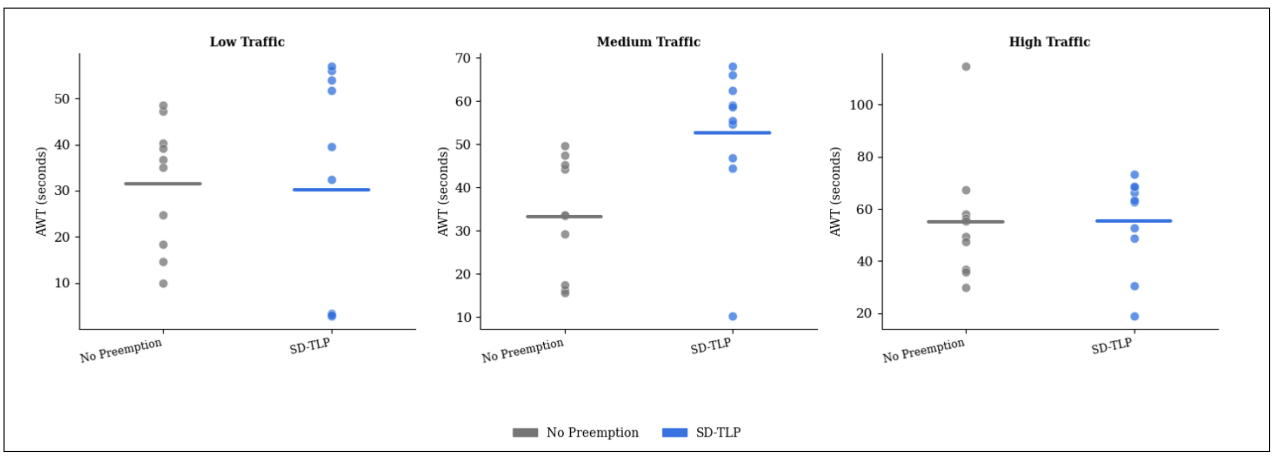


Figure 19: Scenario 1 per-seed scatter plot of AWT, Dots= Individual Seeds, Horizontal Line= Mean

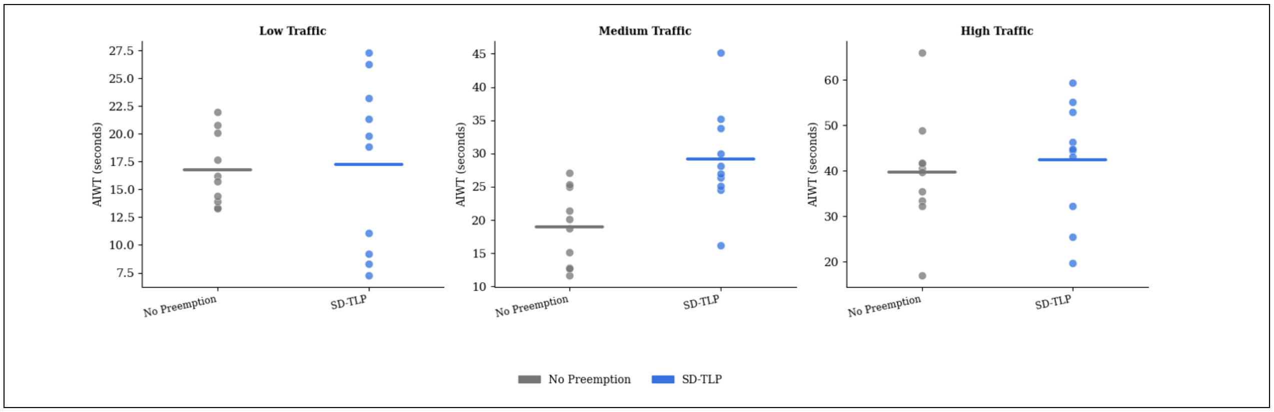


Figure 20: Scenario 1 per-seed scatter plot of AIWT, Dots= Individual Seeds, Horizontal Line= Mean

### Scenario 2: Two Emergency Vehicles (Different Priority)

Figures 21 and 22 show the per seed scatter plots of the average waiting time (AWT) and average imposed waiting time (AIWT) of normal vehicles in scenario 2, respectively.

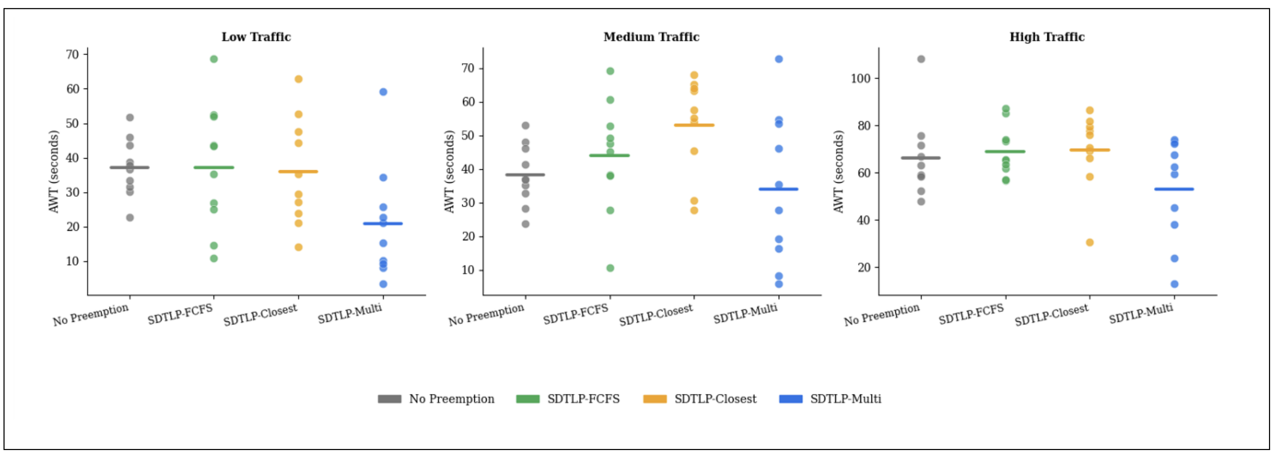


Figure 21: Scenario 2 per-seed scatter plot of AWT, Dots= Individual Seeds, Horizontal Line= Mean

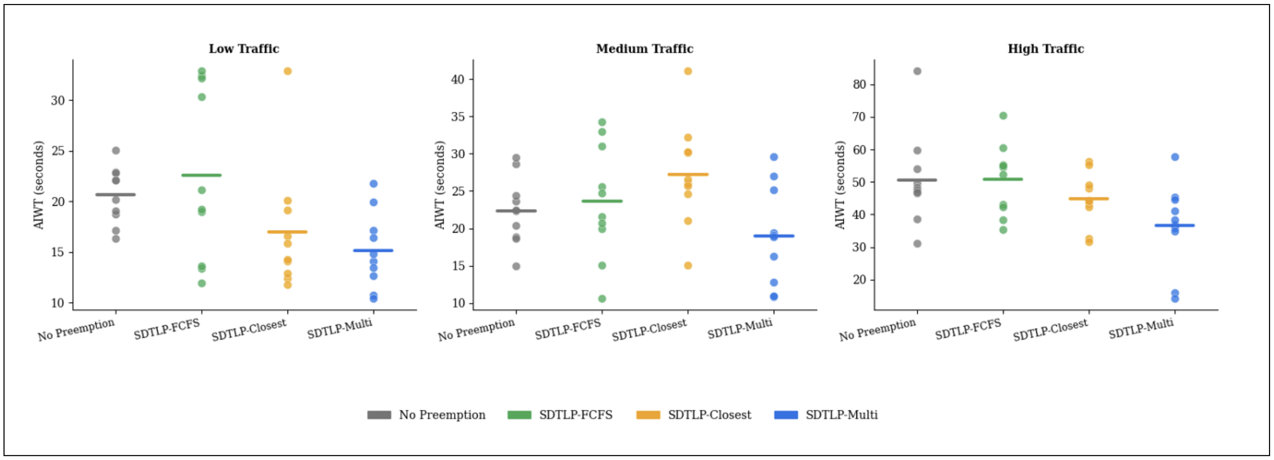


Figure 22: Scenario 2 per-seed scatter plot of AIWT, Dots= Individual Seeds, Horizontal Line= Mean

### Scenario 3: Two Emergency Vehicles (Same Priority)

Figures 23 and 24 show the per seed scatter plots of the average waiting time (AWT) and average imposed waiting time (AIWT) of normal vehicles in scenario 3, respectively.

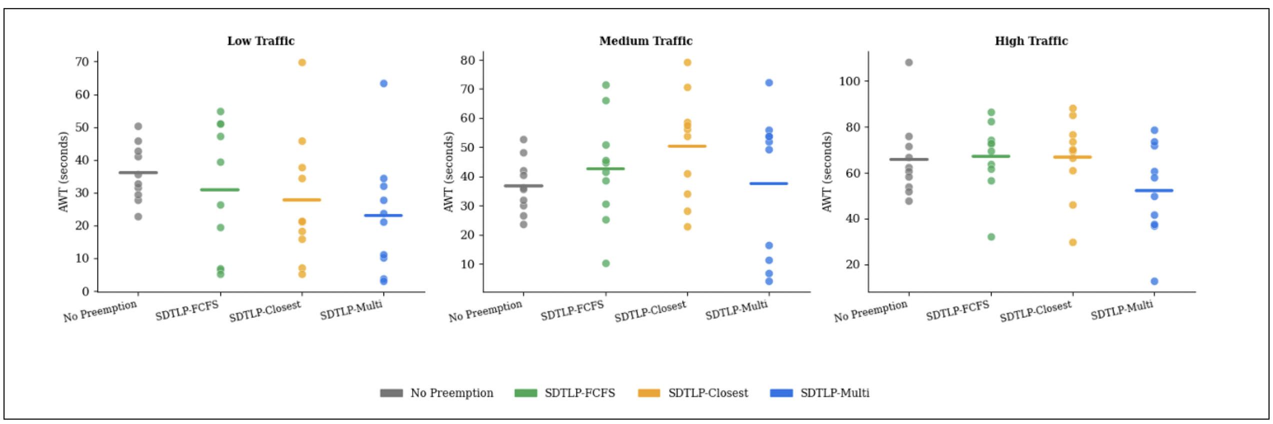


Figure 23: Scenario 3 per-seed scatter plot of AWT, Dots= Individual Seeds, Horizontal Line= Mean

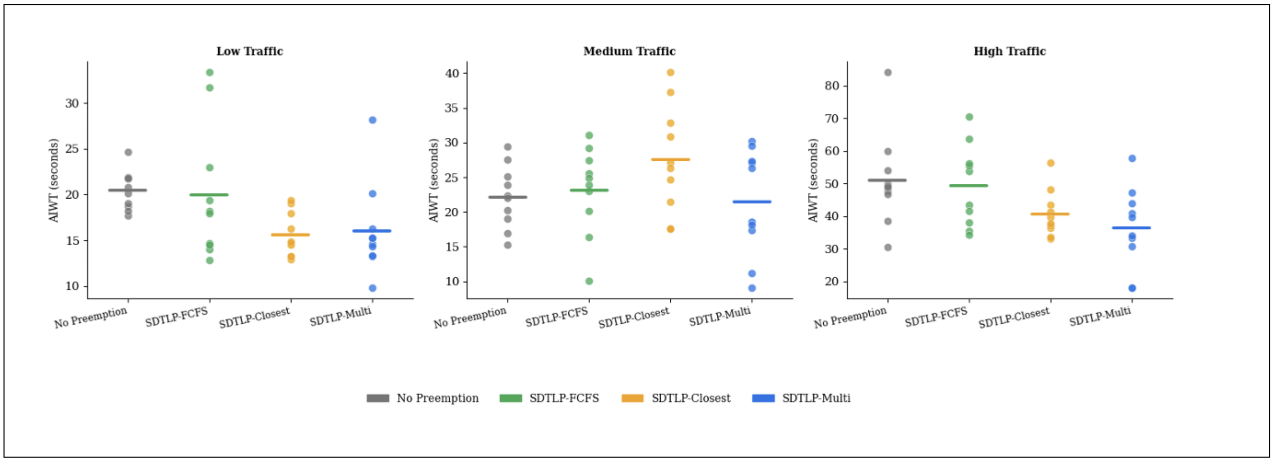


Figure 24: Scenario 3 per-seed scatter plot of AIWT, Dots= Individual Seeds, Horizontal Line= Mean

Figures 25 and 26 show the per seed scatter plots of the average waiting time (AWT) and average imposed waiting time (AIWT) of normal vehicles in scenario 4, respectively.

**Scenario 4: Three Emergency Vehicles**

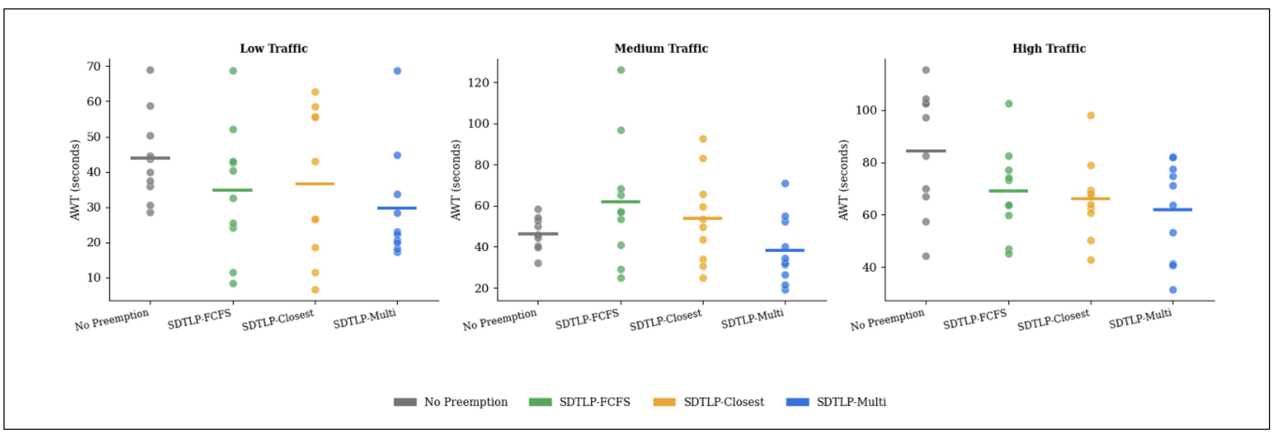


Figure 25: Scenario 4 per-seed scatter plot of AWT, Dots= Individual Seeds, Horizontal Line= Mean

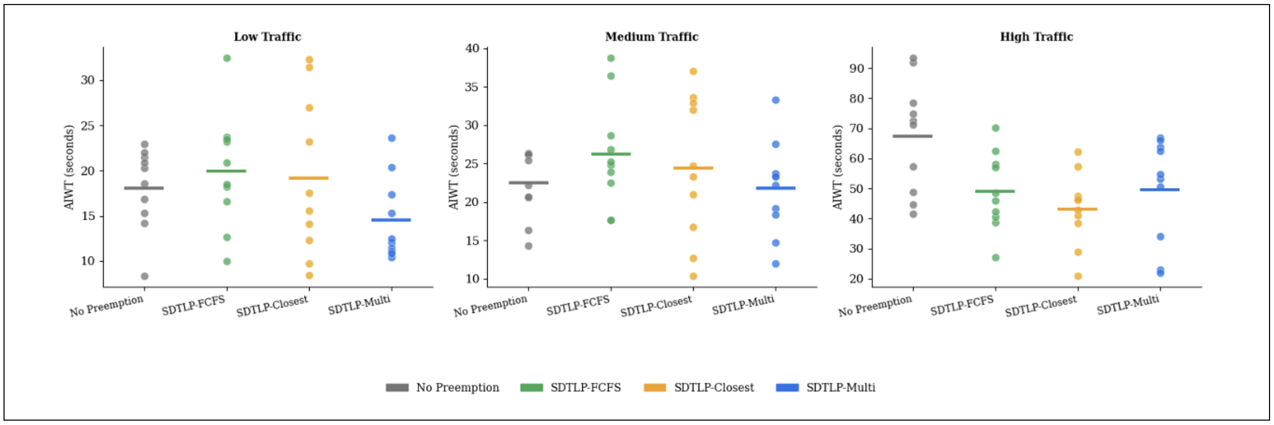


Figure 26: Scenario 4 per-seed scatter plot of AIWT, Dots= Individual Seeds, Horizontal Line= Mean

### Mann–Whitney U Test Results

In this section, the full Mann-Whitney U test results are displayed, as shown in Figures 18,19,20,21.

### Scenario 1: Single Emergency Vehicle

Table 18: Scenario 1: Full Mann–Whitney U Test Results

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | ART | SDTLP vs No Preemption | 21.0 | 0.031019 | TRUE |
| Medium | ART | SDTLP vs No Preemption | 24.0 | 0.053812 | FALSE |
| High | ART | SDTLP vs No Preemption | 8.0 | 0.001699 | TRUE |
| Low | AWT | SDTLP vs No Preemption | 51.0 | 0.969850 | FALSE |
| Medium | AWT | SDTLP vs No Preemption | 85.0 | 0.009108 | TRUE |
| High | AWT | SDTLP vs No Preemption | 61.0 | 0.427355 | FALSE |
| Low | AIWT | SDTLP vs No Preemption | 53.0 | 0.850107 | FALSE |
| Medium | AIWT | SDTLP vs No Preemption | 87.0 | 0.005795 | TRUE |
| High | AIWT | SDTLP vs No Preemption | 63.0 | 0.344704 | FALSE |

### Scenario 2: Two Emergency Vehicles (Different Priority)

Table 19: Scenario 2: Full Mann–Whitney U Test Results

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | ART | SDTLP-Multi vs SDTLP-FCFS | 31.5 | 0.173132 | FALSE |
| Low | ART | SDTLP-Multi vs SDTLP-Closest | 33.0 | 0.211433 | FALSE |
| Low | ART | SDTLP-Multi vs No Preemption | 0.0 | 0.000182 | TRUE |
| Low | ART | SDTLP-FCFS vs SDTLP-Closest | 50.5 | 1.0 | FALSE |
| Low | ART | SDTLP-FCFS vs No Preemption | 0.0 | 0.000182 | TRUE |
| Low | ART | SDTLP-Closest vs No Preemption | 14.0 | 0.007263 | TRUE |
| Medium | ART | SDTLP-Multi vs SDTLP-FCFS | 14.0 | 0.007285 | TRUE |
| Medium | ART | SDTLP-Multi vs SDTLP-Closest | 25.0 | 0.064022 | FALSE |
| Medium | ART | SDTLP-Multi vs No Preemption | 0.0 | 0.000183 | TRUE |
| Medium | ART | SDTLP-FCFS vs SDTLP-Closest | 61.5 | 0.405502 | FALSE |
| Medium | ART | SDTLP-FCFS vs No Preemption | 26.0 | 0.075662 | FALSE |
| Medium | ART | SDTLP-Closest vs No Preemption | 31.0 | 0.161972 | FALSE |
| High | ART | SDTLP-Multi vs SDTLP-FCFS | 21.0 | 0.031209 | TRUE |
| High | ART | SDTLP-Multi vs SDTLP-Closest | 23.0 | 0.045074 | TRUE |
| High | ART | SDTLP-Multi vs No Preemption | 7.0 | 0.001315 | TRUE |
| High | ART | SDTLP-FCFS vs SDTLP-Closest | 61.0 | 0.427181 | FALSE |
| High | ART | SDTLP-FCFS vs No Preemption | 32.0 | 0.185877 | FALSE |
| High | ART | SDTLP-Closest vs No Preemption | 14.0 | 0.007263 | TRUE |
| Low | AWT | SDTLP-Multi vs SDTLP-FCFS | 22.0 | 0.037635 | TRUE |
| Low | AWT | SDTLP-Multi vs SDTLP-Closest | 21.5 | 0.034226 | TRUE |
| Low | AWT | SDTLP-Multi vs No Preemption | 15.0 | 0.009108 | TRUE |
| Low | AWT | SDTLP-FCFS vs SDTLP-Closest | 50.5 | 1.0 | FALSE |
| Low | AWT | SDTLP-FCFS vs No Preemption | 50.0 | 1.0 | FALSE |
| Low | AWT | SDTLP-Closest vs No Preemption | 44.0 | 0.677585 | FALSE |
| Medium | AWT | SDTLP-Multi vs SDTLP-FCFS | 37.0 | 0.344704 | FALSE |
| Medium | AWT | SDTLP-Multi vs SDTLP-Closest | 23.0 | 0.045155 | TRUE |
| Medium | AWT | SDTLP-Multi vs No Preemption | 42.0 | 0.57075 | FALSE |
| Medium | AWT | SDTLP-FCFS vs SDTLP-Closest | 31.5 | 0.173456 | FALSE |
| Medium | AWT | SDTLP-FCFS vs No Preemption | 66.0 | 0.241322 | FALSE |
| Medium | AWT | SDTLP-Closest vs No Preemption | 80.0 | 0.025748 | TRUE |
| High | AWT | SDTLP-Multi vs SDTLP-FCFS | 30.0 | 0.140465 | FALSE |
| High | AWT | SDTLP-Multi vs SDTLP-Closest | 24.0 | 0.053903 | FALSE |
| High | AWT | SDTLP-Multi vs No Preemption | 41.0 | 0.520523 | FALSE |
| High | AWT | SDTLP-FCFS vs SDTLP-Closest | 39.0 | 0.427355 | FALSE |
| High | AWT | SDTLP-FCFS vs No Preemption | 61.0 | 0.427355 | FALSE |
| High | AWT | SDTLP-Closest vs No Preemption | 67.0 | 0.212294 | FALSE |
| Low | AIWT | SDTLP-Multi vs SDTLP-FCFS | 26.0 | 0.075662 | FALSE |
| Low | AIWT | SDTLP-Multi vs SDTLP-Closest | 43.5 | 0.650025 | FALSE |
| Low | AIWT | SDTLP-Multi vs No Preemption | 12.0 | 0.004586 | TRUE |

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | AIWT | SDTLP-FCFS vs SDTLP-Closest | 67.5 | 0.198596 | FALSE |
| Low | AIWT | SDTLP-FCFS vs No Preemption | 52.0 | 0.909722 | FALSE |
| Low | AIWT | SDTLP-Closest vs No Preemption | 19.0 | 0.021134 | TRUE |
| Medium | AIWT | SDTLP-Multi vs SDTLP-FCFS | 31.0 | 0.161972 | FALSE |
| Medium | AIWT | SDTLP-Multi vs SDTLP-Closest | 19.0 | 0.021134 | TRUE |
| Medium | AIWT | SDTLP-Multi vs No Preemption | 36.0 | 0.307489 | FALSE |
| Medium | AIWT | SDTLP-FCFS vs SDTLP-Closest | 36.5 | 0.325569 | FALSE |
| Medium | AIWT | SDTLP-FCFS vs No Preemption | 58.0 | 0.57075 | FALSE |
| Medium | AIWT | SDTLP-Closest vs No Preemption | 77.0 | 0.045155 | TRUE |
| High | AIWT | SDTLP-Multi vs SDTLP-FCFS | 22.0 | 0.037635 | TRUE |
| High | AIWT | SDTLP-Multi vs SDTLP-Closest | 32.0 | 0.185877 | FALSE |
| High | AIWT | SDTLP-Multi vs No Preemption | 18.0 | 0.017257 | TRUE |
| High | AIWT | SDTLP-FCFS vs SDTLP-Closest | 62.0 | 0.384673 | FALSE |
| High | AIWT | SDTLP-FCFS vs No Preemption | 55.0 | 0.73373 | FALSE |
| High | AIWT | SDTLP-Closest vs No Preemption | 37.0 | 0.344704 | FALSE |

### Scenario 3: Two Emergency Vehicles (Same Priority)

Table 20: Scenario 3: Full Mann–Whitney U Test Results

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | ART | SDTLP-Multi vs SDTLP-FCFS | 27.0 | 0.088853 | FALSE |
| Low | ART | SDTLP-Multi vs SDTLP-Closest | 27.5 | 0.095803 | FALSE |
| Low | ART | SDTLP-Multi vs No Preemption | 1.0 | 0.000246 | TRUE |
| Low | ART | SDTLP-FCFS vs SDTLP-Closest | 52.5 | 0.879694 | FALSE |
| Low | ART | SDTLP-FCFS vs No Preemption | 7.0 | 0.001309 | TRUE |
| Low | ART | SDTLP-Closest vs No Preemption | 20.0 | 0.025692 | TRUE |
| Medium | ART | SDTLP-Multi vs SDTLP-FCFS | 13.0 | 0.005795 | TRUE |
| Medium | ART | SDTLP-Multi vs SDTLP-Closest | 22.0 | 0.037492 | TRUE |
| Medium | ART | SDTLP-Multi vs No Preemption | 4.0 | 0.000583 | TRUE |
| Medium | ART | SDTLP-FCFS vs SDTLP-Closest | 63.0 | 0.344704 | FALSE |
| Medium | ART | SDTLP-FCFS vs No Preemption | 35.0 | 0.273036 | FALSE |
| Medium | ART | SDTLP-Closest vs No Preemption | 29.5 | 0.130425 | FALSE |
| High | ART | SDTLP-Multi vs SDTLP-FCFS | 16.0 | 0.01133 | TRUE |
| High | ART | SDTLP-Multi vs SDTLP-Closest | 22.0 | 0.037635 | TRUE |
| High | ART | SDTLP-Multi vs No Preemption | 7.0 | 0.001315 | TRUE |
| High | ART | SDTLP-FCFS vs SDTLP-Closest | 70.5 | 0.130425 | FALSE |
| High | ART | SDTLP-FCFS vs No Preemption | 35.0 | 0.273036 | FALSE |
| High | ART | SDTLP-Closest vs No Preemption | 13.0 | 0.005795 | TRUE |
| Low | AWT | SDTLP-Multi vs SDTLP-FCFS | 39.0 | 0.427355 | FALSE |

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | AWT | SDTLP-Multi vs SDTLP-Closest | 42.0 | 0.57046 | FALSE |
| Low | AWT | SDTLP-Multi vs No Preemption | 22.0 | 0.037635 | TRUE |
| Low | AWT | SDTLP-FCFS vs SDTLP-Closest | 56.0 | 0.677585 | FALSE |
| Low | AWT | SDTLP-FCFS vs No Preemption | 46.0 | 0.791337 | FALSE |
| Low | AWT | SDTLP-Closest vs No Preemption | 29.0 | 0.121225 | FALSE |
| Medium | AWT | SDTLP-Multi vs SDTLP-FCFS | 51.0 | 0.96985 | FALSE |
| Medium | AWT | SDTLP-Multi vs SDTLP-Closest | 31.5 | 0.173456 | FALSE |
| Medium | AWT | SDTLP-Multi vs No Preemption | 58.0 | 0.57075 | FALSE |
| Medium | AWT | SDTLP-FCFS vs SDTLP-Closest | 39.0 | 0.427355 | FALSE |
| Medium | AWT | SDTLP-FCFS vs No Preemption | 62.0 | 0.384673 | FALSE |
| Medium | AWT | SDTLP-Closest vs No Preemption | 73.0 | 0.088973 | FALSE |
| High | AWT | SDTLP-Multi vs SDTLP-FCFS | 28.0 | 0.10411 | FALSE |
| High | AWT | SDTLP-Multi vs SDTLP-Closest | 29.0 | 0.121225 | FALSE |
| High | AWT | SDTLP-Multi vs No Preemption | 33.0 | 0.212294 | FALSE |
| High | AWT | SDTLP-FCFS vs SDTLP-Closest | 50.0 | 1.0 | FALSE |
| High | AWT | SDTLP-FCFS vs No Preemption | 63.0 | 0.344704 | FALSE |
| High | AWT | SDTLP-Closest vs No Preemption | 60.0 | 0.472676 | FALSE |
| Low | AIWT | SDTLP-Multi vs SDTLP-FCFS | 34.0 | 0.241322 | FALSE |
| Low | AIWT | SDTLP-Multi vs SDTLP-Closest | 50.0 | 1.0 | FALSE |
| Low | AIWT | SDTLP-Multi vs No Preemption | 14.0 | 0.007285 | TRUE |
| Low | AIWT | SDTLP-FCFS vs SDTLP-Closest | 66.0 | 0.241322 | FALSE |
| Low | AIWT | SDTLP-FCFS vs No Preemption | 36.0 | 0.307489 | FALSE |
| Low | AIWT | SDTLP-Closest vs No Preemption | 9.0 | 0.002202 | TRUE |
| Medium | AIWT | SDTLP-Multi vs SDTLP-FCFS | 46.0 | 0.791337 | FALSE |
| Medium | AIWT | SDTLP-Multi vs SDTLP-Closest | 32.5 | 0.198596 | FALSE |
| Medium | AIWT | SDTLP-Multi vs No Preemption | 50.0 | 1.0 | FALSE |
| Medium | AIWT | SDTLP-FCFS vs SDTLP-Closest | 35.0 | 0.273036 | FALSE |
| Medium | AIWT | SDTLP-FCFS vs No Preemption | 59.0 | 0.520523 | FALSE |
| Medium | AIWT | SDTLP-Closest vs No Preemption | 71.0 | 0.121225 | FALSE |
| High | AIWT | SDTLP-Multi vs SDTLP-FCFS | 24.0 | 0.053903 | FALSE |
| High | AIWT | SDTLP-Multi vs SDTLP-Closest | 41.0 | 0.520523 | FALSE |
| High | AIWT | SDTLP-Multi vs No Preemption | 20.0 | 0.025748 | TRUE |
| High | AIWT | SDTLP-FCFS vs SDTLP-Closest | 48.0 | 0.909722 | FALSE |
| High | AIWT | SDTLP-FCFS vs No Preemption | 23.0 | 0.045155 | TRUE |
| High | AIWT | SDTLP-Closest vs No Preemption | 71.0 | 0.121225 | FALSE |

### Scenario 4: Three Emergency Vehicles

Table 21: Scenario 4: Full Mann–Whitney U Test Results

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | ART | SDTLP-Multi vs SDTLP-FCFS | 20.5 | 0.028306 | TRUE |
| Low | ART | SDTLP-Multi vs SDTLP-Closest | 35.0 | 0.273036 | FALSE |
| Low | ART | SDTLP-Multi vs No Preemption | 0.0 | 0.000183 | TRUE |
| Low | ART | SDTLP-FCFS vs SDTLP-Closest | 55.0 | 0.733730 | FALSE |
| Low | ART | SDTLP-FCFS vs No Preemption | 12.0 | 0.004586 | TRUE |
| Low | ART | SDTLP-Closest vs No Preemption | 16.0 | 0.011330 | TRUE |
| Medium | ART | SDTLP-Multi vs SDTLP-FCFS | 26.0 | 0.075442 | FALSE |
| Medium | ART | SDTLP-Multi vs SDTLP-Closest | 51.0 | 0.969850 | FALSE |
| Medium | ART | SDTLP-Multi vs No Preemption | 7.0 | 0.001315 | TRUE |
| Medium | ART | SDTLP-FCFS vs SDTLP-Closest | 76.0 | 0.053903 | FALSE |
| Medium | ART | SDTLP-FCFS vs No Preemption | 16.5 | 0.012578 | TRUE |
| Medium | ART | SDTLP-Closest vs No Preemption | 6.0 | 0.001008 | TRUE |
| High | ART | SDTLP-Multi vs SDTLP-FCFS | 42.0 | 0.570750 | FALSE |
| High | ART | SDTLP-Multi vs SDTLP-Closest | 59.0 | 0.520523 | FALSE |
| High | ART | SDTLP-Multi vs No Preemption | 2.0 | 0.000330 | TRUE |
| High | ART | SDTLP-FCFS vs SDTLP-Closest | 64.0 | 0.307489 | FALSE |
| High | ART | SDTLP-FCFS vs No Preemption | 19.0 | 0.021134 | TRUE |
| High | ART | SDTLP-Closest vs No Preemption | 7.0 | 0.001315 | TRUE |
| Low | AWT | SDTLP-Multi vs SDTLP-FCFS | 38.0 | 0.384673 | FALSE |
| Low | AWT | SDTLP-Multi vs SDTLP-Closest | 42.0 | 0.570750 | FALSE |
| Low | AWT | SDTLP-Multi vs No Preemption | 18.0 | 0.017257 | TRUE |
| Low | AWT | SDTLP-FCFS vs SDTLP-Closest | 45.0 | 0.733730 | FALSE |
| Low | AWT | SDTLP-FCFS vs No Preemption | 34.0 | 0.241322 | FALSE |
| Low | AWT | SDTLP-Closest vs No Preemption | 38.0 | 0.384673 | FALSE |
| Medium | AWT | SDTLP-Multi vs SDTLP-FCFS | 24.0 | 0.053903 | FALSE |
| Medium | AWT | SDTLP-Multi vs SDTLP-Closest | 30.0 | 0.140465 | FALSE |
| Medium | AWT | SDTLP-Multi vs No Preemption | 30.0 | 0.140465 | FALSE |
| Medium | AWT | SDTLP-FCFS vs SDTLP-Closest | 58.0 | 0.570750 | FALSE |
| Medium | AWT | SDTLP-FCFS vs No Preemption | 69.0 | 0.161972 | FALSE |
| Medium | AWT | SDTLP-Closest vs No Preemption | 58.0 | 0.570750 | FALSE |
| High | AWT | SDTLP-Multi vs SDTLP-FCFS | 41.0 | 0.520523 | FALSE |
| High | AWT | SDTLP-Multi vs SDTLP-Closest | 48.0 | 0.909722 | FALSE |
| High | AWT | SDTLP-Multi vs No Preemption | 23.0 | 0.045155 | TRUE |
| High | AWT | SDTLP-FCFS vs SDTLP-Closest | 55.0 | 0.733730 | FALSE |
| High | AWT | SDTLP-FCFS vs No Preemption | 31.0 | 0.161972 | FALSE |
| High | AWT | SDTLP-Closest vs No Preemption | 25.0 | 0.064022 | FALSE |
| Low | AIWT | SDTLP-Multi vs SDTLP-FCFS | 24.0 | 0.053903 | FALSE |
| Low | AIWT | SDTLP-Multi vs SDTLP-Closest | 35.0 | 0.273036 | FALSE |
| Low | AIWT | SDTLP-Multi vs No Preemption | 28.0 | 0.104110 | FALSE |

| Traffic | Metric | Comparison | U-stat | p-value | Significant |
|---|---|---|---|---|---|
| Low | AIWT | SDTLP-FCFS vs SDTLP-Closest | 59.0 | 0.520523 | FALSE |
| Low | AIWT | SDTLP-FCFS vs No Preemption | 60.0 | 0.472676 | FALSE |
| Low | AIWT | SDTLP-Closest vs No Preemption | 51.0 | 0.969850 | FALSE |
| Medium | AIWT | SDTLP-Multi vs SDTLP-FCFS | 30.0 | 0.140465 | FALSE |
| Medium | AIWT | SDTLP-Multi vs SDTLP-Closest | 41.0 | 0.520523 | FALSE |
| Medium | AIWT | SDTLP-Multi vs No Preemption | 45.0 | 0.733730 | FALSE |
| Medium | AIWT | SDTLP-FCFS vs SDTLP-Closest | 58.0 | 0.570750 | FALSE |
| Medium | AIWT | SDTLP-FCFS vs No Preemption | 64.0 | 0.307489 | FALSE |
| Medium | AIWT | SDTLP-Closest vs No Preemption | 56.0 | 0.677585 | FALSE |
| High | AIWT | SDTLP-Multi vs SDTLP-FCFS | 54.0 | 0.791337 | FALSE |
| High | AIWT | SDTLP-Multi vs SDTLP-Closest | 68.0 | 0.185877 | FALSE |
| High | AIWT | SDTLP-Multi vs No Preemption | 25.0 | 0.064022 | FALSE |
| High | AIWT | SDTLP-FCFS vs SDTLP-Closest | 61.0 | 0.427355 | FALSE |
| High | AIWT | SDTLP-FCFS vs No Preemption | 20.0 | 0.025748 | TRUE |
| High | AIWT | SDTLP-Closest vs No Preemption | 15.0 | 0.009108 | TRUE |